Name:____________

Double Award Science – Biology Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	3	7	14	21	28	35	42
2018	7	9	13	20	27	34	41
2019	5	8	15	23	31	39	47
2022	5	7	14	22	30	38	46
2023	6	9	13	19	26	33	40

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3430UA0-1

S17-3430UA0-1

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
HIGHER TIER

WEDNESDAY, 14 JUNE 2017 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**10**	
3.	**6**	
4.	**8**	
5.	**8**	
6.	**10**	
7.	**7**	
8.	**6**	
Total	**60**	

3430UA01
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

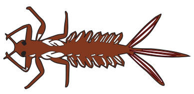
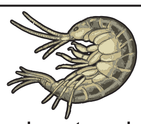
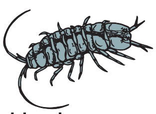
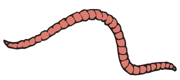
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

Answer **all** *questions.*

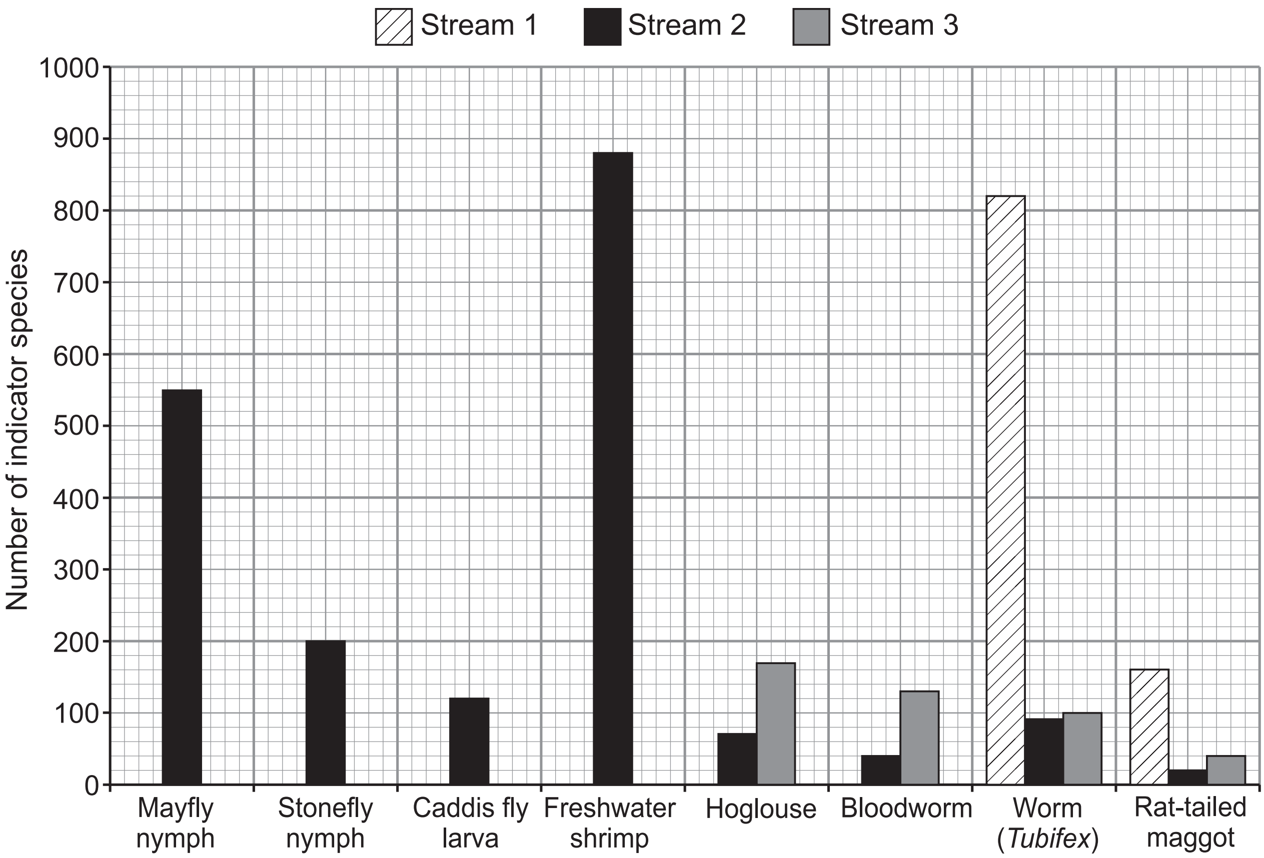
1.

Following flooding in parts of Wales in 2015, environmental scientists investigated nitrate pollution in three different streams. They counted the numbers of eight different indicator species in equal lengths of the three streams, on the same day. The indicator species they counted are shown in the chart below.

GROUP	INDICATOR SPECIES		POLLUTION LEVEL
A	 Mayfly nymph	 Stonefly nymph	If you find these animals in your stream, then there is **LITTLE OR NO POLLUTION**
B	 Caddis fly larva	 Freshwater shrimp	If you find these animals, but none from Group **A**, then there may be **SLIGHT POLLUTION**
C	 Hoglouse	 Bloodworm	If you find these animals, but none from Groups **A** or **B**, then there is probably **MEDIUM POLLUTION**
D	 Worm (*Tubifex*)	 Rat-tailed maggot	If you find these animals, but none from Groups **A**, **B** or **C**, then there is **HIGH POLLUTION**
E	No live animals found		If you find no animals at all, then there is **VERY HIGH POLLUTION**

Examiner only

Graph showing the number of indicator species found in each of the streams 1, 2 and 3



(a) (i) Using the graph above and the chart on the opposite page state the pollution level in each of the three streams. [2]

stream **1** ....

stream **2** ....

stream **3** ....

(ii) Suggest **two** possible **sources** of nitrate pollution in streams. [2]

....

....

(b) Name **one** type of indicator species which could be used to assess levels of air pollution. [1]

....

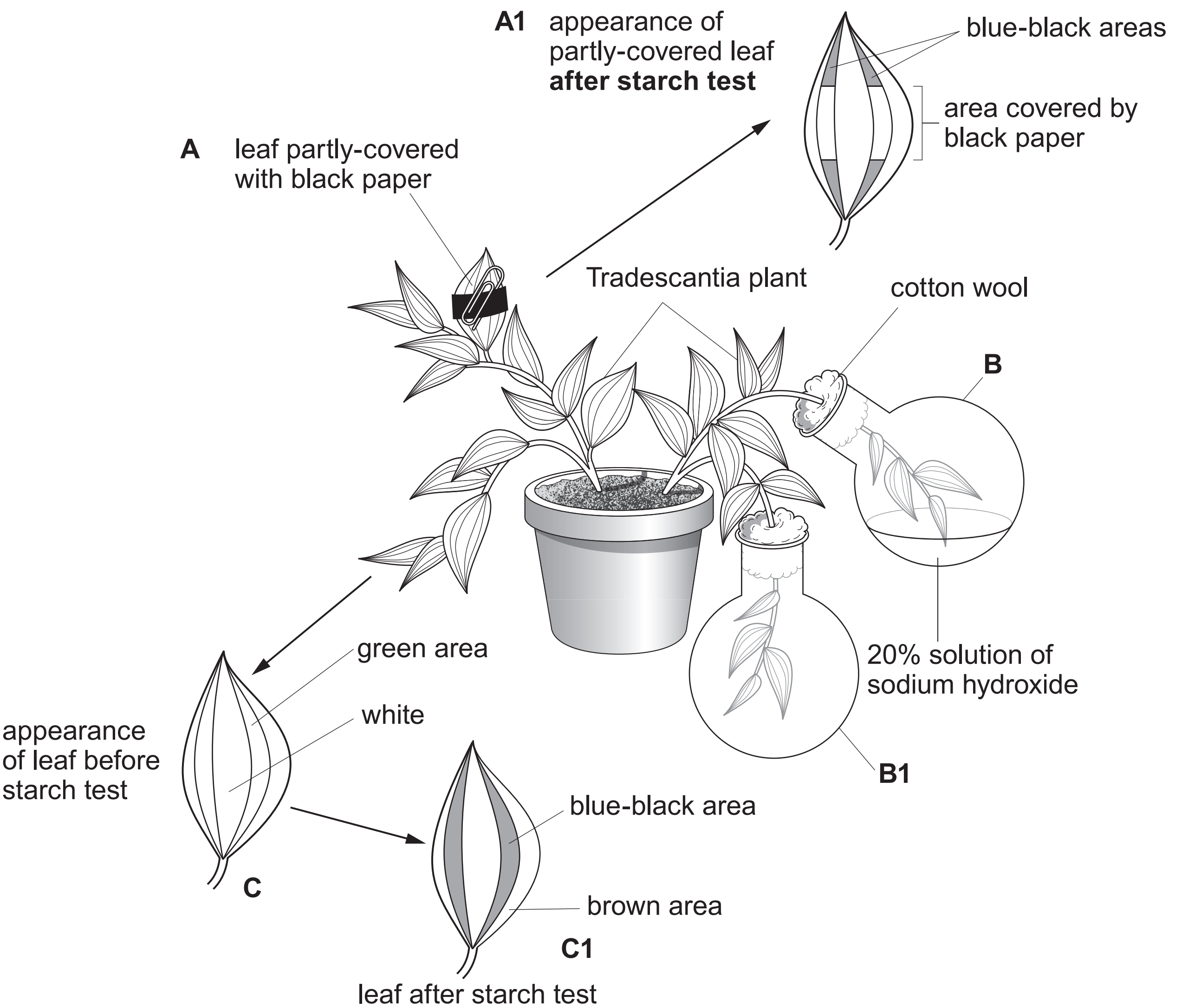
5

3430UA01 03

2. *Tradescantia* is a plant whose leaves have green and white areas.

Tradescantia leaves

In the experiment below, a *Tradescantia* plant is used in an investigation to demonstrate **three** factors, **A**, **B** and **C**, needed for photosynthesis.



Examiner only

(a) (i) State the factor needed for photosynthesis, which is being demonstrated in each of the following: [3]

A ....

B ....

C ....

(ii) How could the experiment in flasks **B** and **B1** be improved? Explain your answer. [2]

....

....

(iii) Before the apparatus was set up, the *Tradescantia* plant was kept in a dark cupboard for 48 hours. Explain the reason for this. [2]

....

....

....

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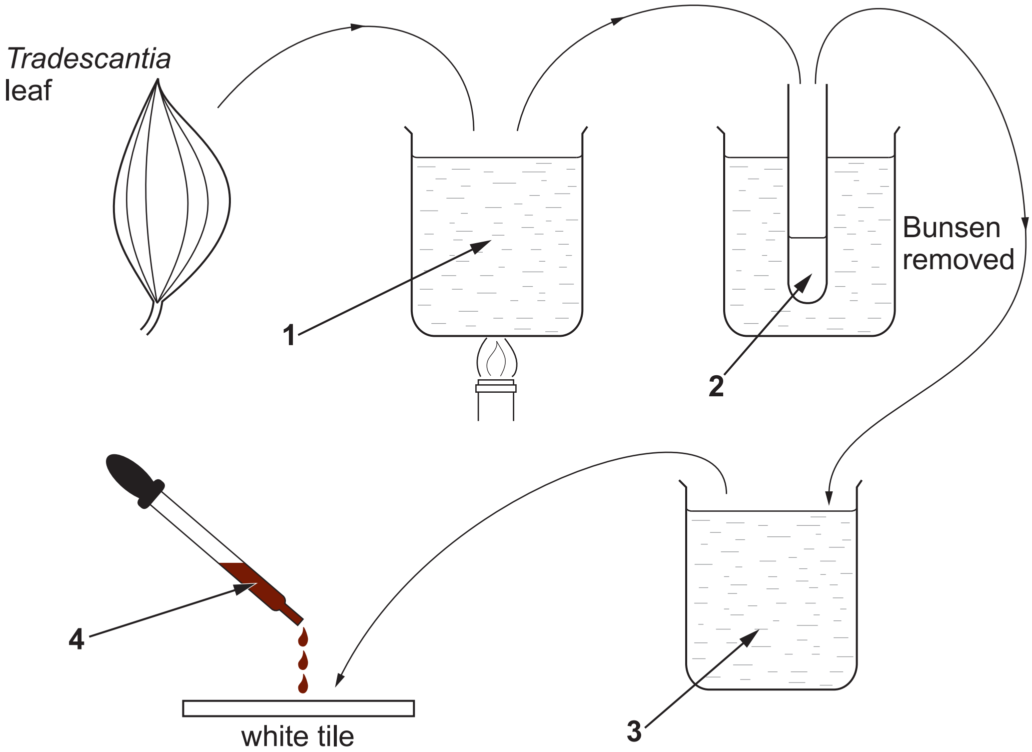
Examiner only

(b) The diagram below shows the stages in testing a leaf for starch.



Name the liquids **1**, **2**, **3 and 4** shown in the diagram. [3]

1 ....

2 ....

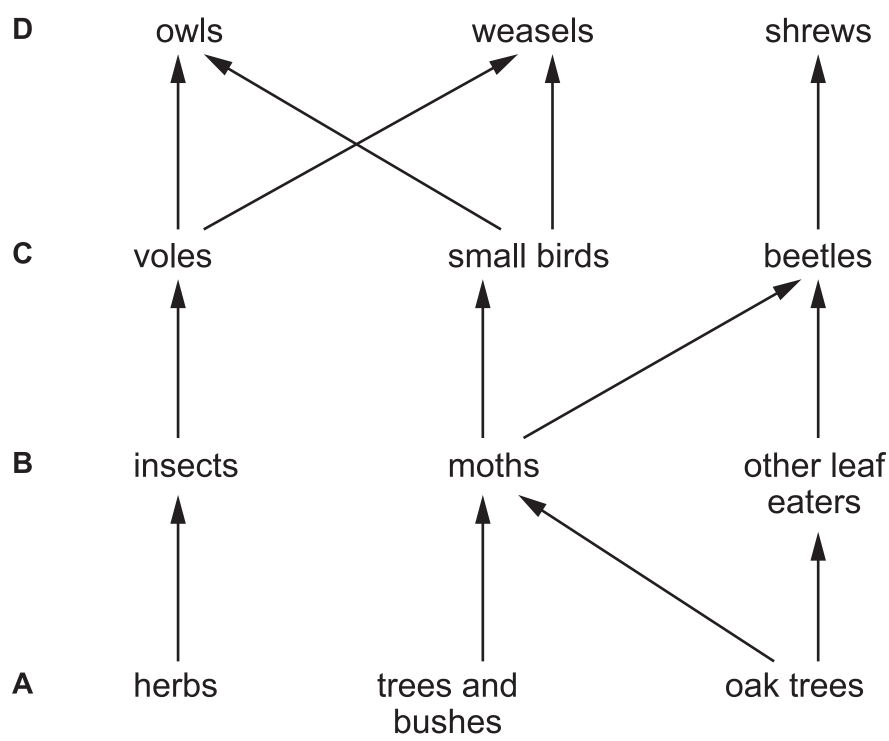
3 ....

4 ....

10

Examiner only

3. The diagram below shows a food web for a woodland.



(a) The food web shows four different trophic levels. Give the term used to describe each of the trophic levels. [1]

A ....

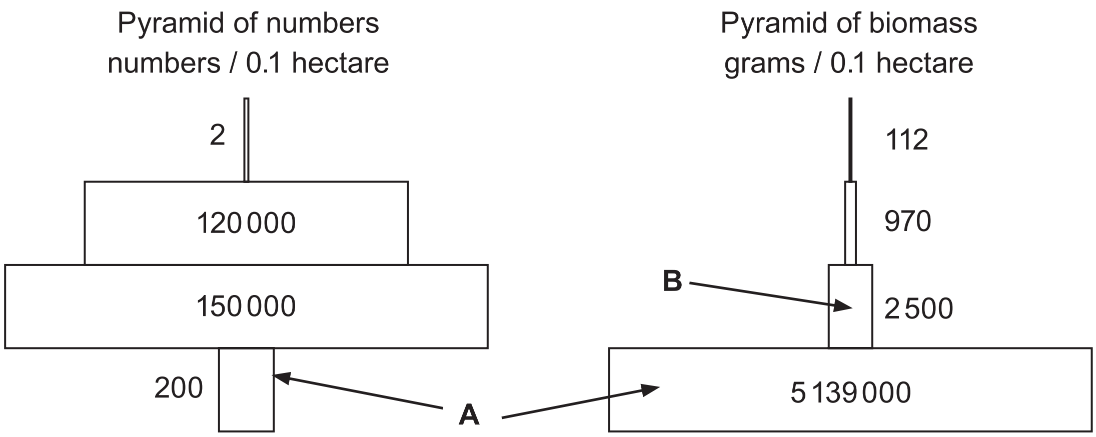
B ....

C ....

D ....

Examiner only

(b) The diagrams below show a pyramid of numbers and a pyramid of biomass for a food chain in this food web.



(i) Calculate the percentage of biomass passed from trophic level **A** to trophic level **B**. [2]

Percentage of biomass = ... %

(ii) Explain why the trophic level labelled **A** is such a different width in the two pyramids. [2]

....

....

....

(iii) State **one** way in which energy is lost between trophic levels. [1]

....

....

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6

Examiner only

4. *(a)* State the meaning of the term diffusion. [1]

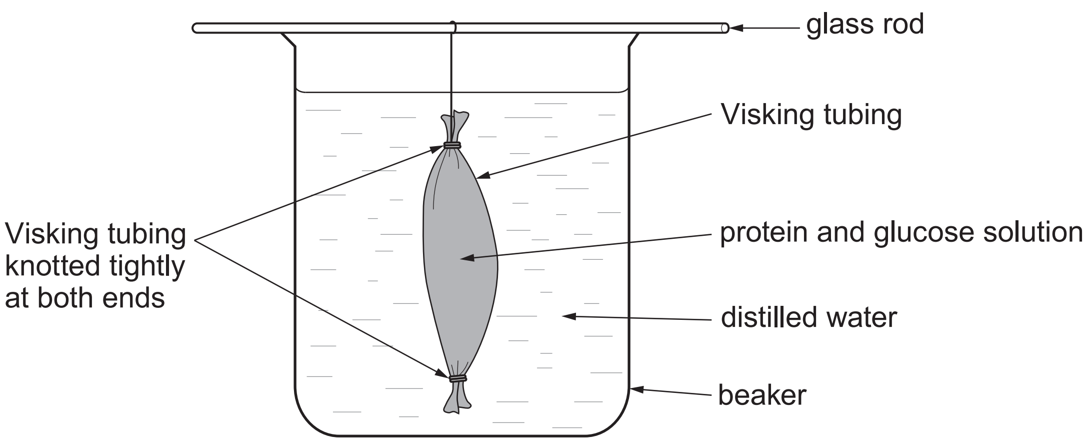
....

....

(b) After a lesson on the properties of cell membranes a year 10 class was asked to investigate some of these properties using Visking tubing. They were given the following instructions:

- Soak a 15 cm length of Visking tubing in water to soften it.
- Tie a knot in one end of the tube.
- Fill the tube with a solution made up of protein and glucose dissolved in water.
- Tie a knot in the open end of the tube.
- Wash the tube under a stream of tap water for 15 seconds.
- Using a glass rod suspend the Visking tubing in a beaker of distilled water.

The diagram below shows how your apparatus should appear.



(i) Why were the students instructed to *'wash the tube under a stream of tap water for 15 seconds'*? [1]

....

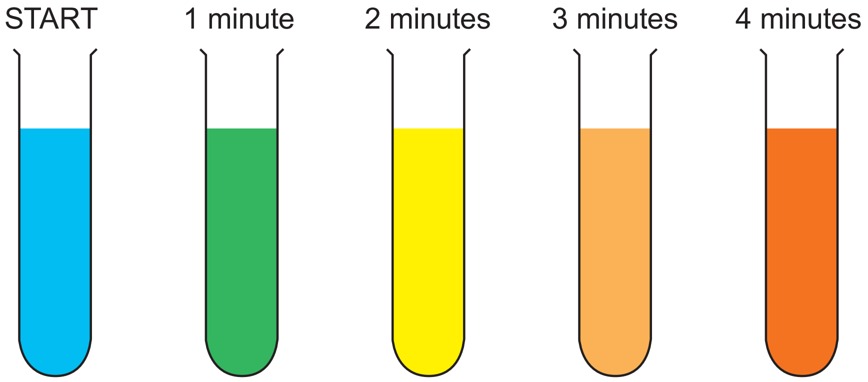
....

 (3430UA0-1)

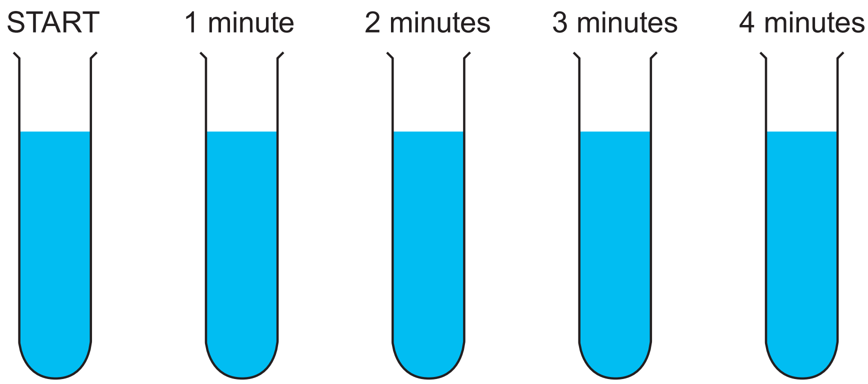
Examiner only

The students were asked to sample the distilled water **in the beaker** for the presence of both glucose and protein at the start of the experiment and every minute for the next four minutes. Dafydd decided to photograph his results on his smart phone. The photographs he obtained are shown below.

Benedict's test for glucose



Biuret test for protein



(ii) Explain Dafydd's results, **at four minutes**, for the contents of the **beaker** for both the Benedict's test and the Biuret test.

Benedict's test [3]

....

....

....

....

Biuret test [3]

....

....

....

....

8

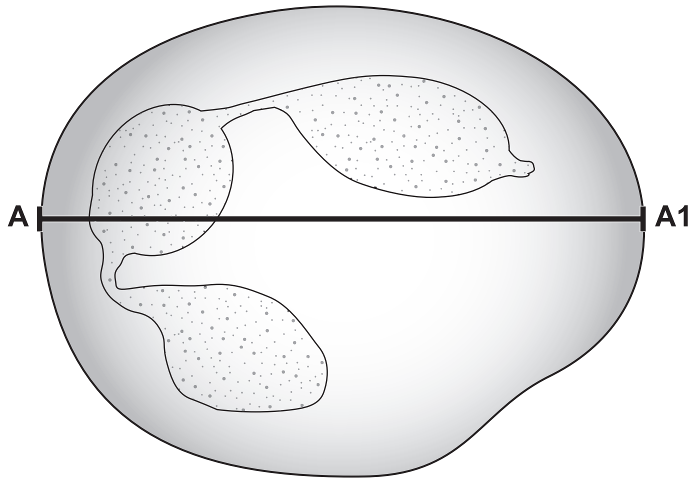
Examiner only

5. *(a)* Sarah was viewing a smear of blood under a light microscope. How could she calculate the magnification of the microscope she was using? [1]

....

....

(b) Sarah produced the drawing below. It shows one of the blood cells she looked at under the microscope.



(i) Name the type of white blood cell shown in the drawing above. [1]

....

(ii) The actual size of the white blood cell, across the line **A – A1,** was 20 μm (20 micrometres). There are 1000 μm in 1 mm.

I. Measure the length of the line **A – A1** in the diagram above. [1]

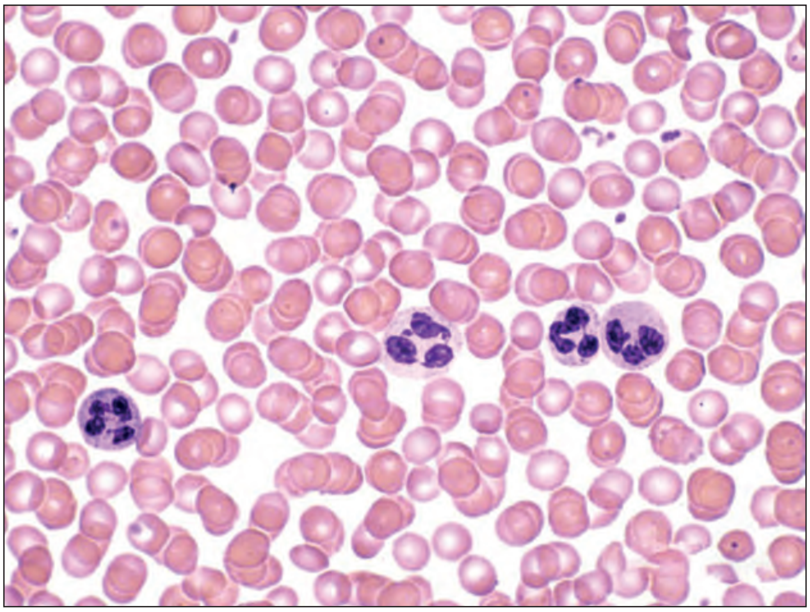
length of the line **A – A1** = ... mm

II. Calculate how many times bigger Sarah's drawing is than the actual blood cell. [2]

Magnification = × ....

Examiner only

(c) The photograph below shows a smear of blood viewed under the high power magnification of a microscope.



(i) Describe the procedure involved in preparing these blood cells for viewing under the microscope and state the reason why this is needed. [2]

....

....

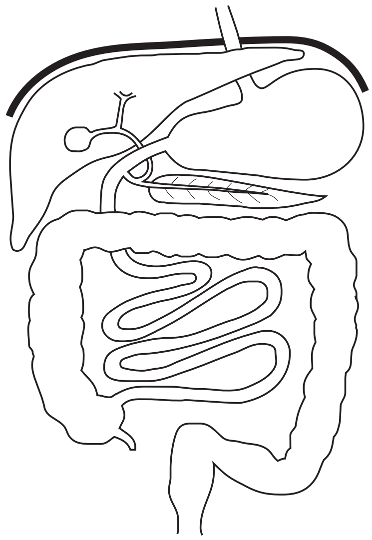
....

(ii) The ratio of red blood cells to white blood cells is 500:1. A sample of blood has 57 000 red blood cells.
Calculate the number of white blood cells in the sample. [1]

Number of white blood cells = ...

8

6. The diagram below shows part of the human digestive system.



(a) **On the diagram above label** the following: [2]

A pancreas
B bile duct

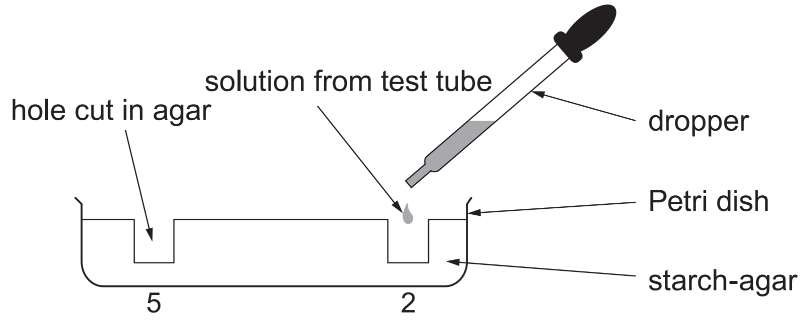
(b) The diagram below shows five test tubes **A – E** and their contents. The pH and temperature of each is also shown.

	Solution				
	Distilled water	0.5% amylase	1% amylase	1% amylase	2% amylase
tube	**A**	**B**	**C**	**D**	**E**
temperature (°C)	37	37	100	37	37
pH	7	7	7	7	7

A pupil carried out the following procedure:

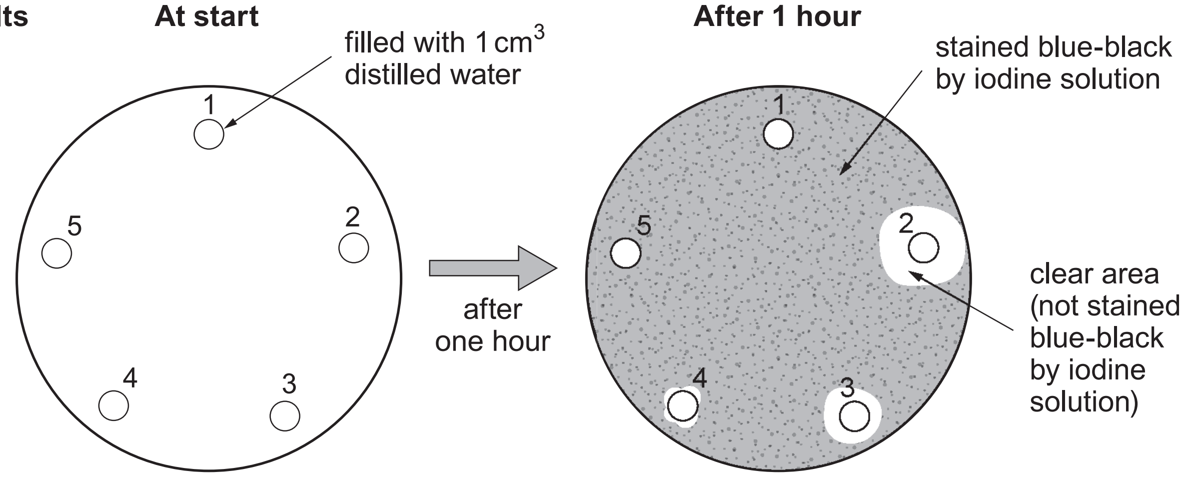
- She took a Petri dish containing starch-agar (a set jelly in which starch had been dissolved).
- She cut five small holes in the jelly (holes 1 – 5).
- She put 1 cm^3 distilled water from Tube **A** into hole 1.
- Holes 2 – 5 were each filled with 1 cm^3 of a different solution from one of test tubes **B**, **C**, **D** or **E**.
- After one hour she flooded the Petri dish with iodine solution.

Examiner only



Cross-sectional diagram showing two of the five holes.

Results



(i) Explain the appearance of the clear areas. [1]

....

....

(ii) State the solution that was added to hole **2**. Explain your answer. [2]

....

....

(iii) Which hole contained the contents of tube **C**? Explain your answer. [4]

Hole number ....

Explanation ....

....

....

....

(c) State the purpose of tube **A**, the contents of which was put into hole **1**. [1]

....

10

Examiner only

7. The heart is a pump which supplies blood to the organs of the body. It deals with increasing physical activity, such as exercise, by increasing its rate (number of beats per minute). After exercise, the heart rate will eventually return to the resting rate. The time the heart takes to return to its resting rate is a good indication of a person's level of fitness.

A researcher wanted to investigate the difference in the level of fitness between men and women. Six people at a gym volunteered to compare their fitness. Each volunteer ran on a treadmill for five minutes. Immediately at the end of the exercise their heart rates were recorded on a laptop computer. Their heart rates were also recorded one minute later.

The researcher conducting the investigation used the data collected from each volunteer to calculate their fitness index. An index of 60 – 70 shows a person who is reasonably fit, below this level shows decreasing fitness.

The results of the investigation are shown in the table below.

Volunteer	Resting heart rate (beats/min)	Heart rate immediately after exercise (beats/min)	Heart rate 1 min after exercise (beats/min)	Fitness index	Body mass (kg)
John	70	190	130	35	56.5
Michael	70	150	80	60	63.6
Charlotte	60	110	65	75	76.4
Paul	80	195	140	20	101.8
Peter	75	180	156	43	59.1
Lucy	70	130	80	68	56.1

(a) (i) Who appears to be the least fit person in the table? Explain your answer. [1]

....

....

 (3430UA0-1)

Examiner only

(ii) What is the relationship between **fitness index** and heart rate immediately after exercise? [1]

....

....

(iii) At the end of the investigation the volunteers discussed the results. Peter said that "he thought that John was fitter than Lucy because one minute after exercise John's heart rate had fallen by 60 beats / min whereas Lucy's had only fallen by 50 beats / min." The researcher said that you must calculate the percentage decrease in the heart rates in order to make a fair comparison.

I. In the space below calculate the percentage decrease in the heart rates of both John and Lucy. [2]

John

Percentage decrease in heart rate = ...

Lucy

Percentage decrease in heart rate = ...

II. From the answers to (I.) above state whether John or Lucy appears fittest. [1]

....

(b) The design of this investigation has many faults.

Give **two** factors which should be controlled in this investigation. [1]

....

....

(c) Why is it not possible to conclude from the data that Paul's low **fitness index** is due to obesity? [1]

....

7

Examiner only

8. The following word equations show the two types of cell respiration which occur in humans.

Equation 1

Glucose + Oxygen ⟶ Carbon dioxide + Water + ATP

Equation 2

Glucose ⟶ Lactic acid + ATP

Name each of the types of cell respiration shown above and write an account explaining when each occurs in the human body. Include any advantages or disadvantages of each type of respiration. [6 QER]

....

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6

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3430UA0-1

S18-3430UA0-1

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
HIGHER TIER

MONDAY, 11 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**9**	
3.	**6**	
4.	**6**	
5.	**9**	
6.	**10**	
7.	**6**	
8.	**8**	
Total	**60**	

3430UA01
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

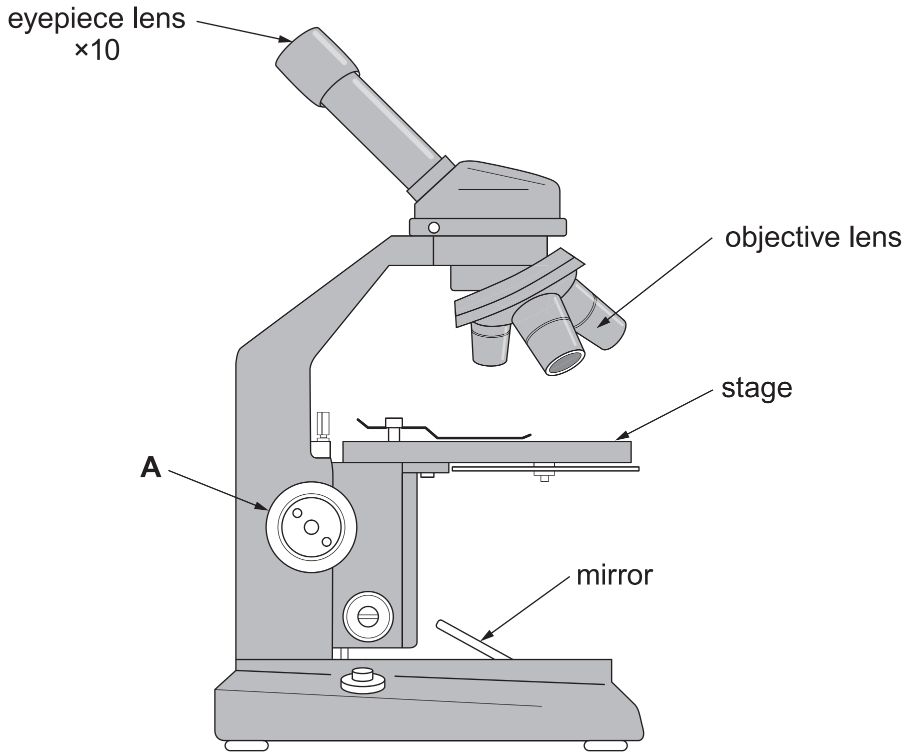
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** *questions.*

1. Rhys was asked by his teacher to set up a light microscope so that he could view some cells at a magnification of ×100. The microscope had three objective lenses of ×4, ×10 and ×40 magnifications. Rhys was also given a prepared slide of muscle cells.



(a) Explain how Rhys could view the muscle cells at a magnification of ×100. [2]

....

....

....

(b) State the function of structure **A** on the diagram. [1]

....

Examiner only

(c) When Rhys viewed the muscle cell under the microscope he could see that the cells were not found on their own, but were grouped together in large numbers.

(i) Muscle cells are described as being specialised cells. State the advantage to the organism of having specialised cells. [1]

....

....

(ii) State the name given to a large number of the same cells grouped together. [1]

....

(d) In biology, what is meant by the term organ? [1]

....

....

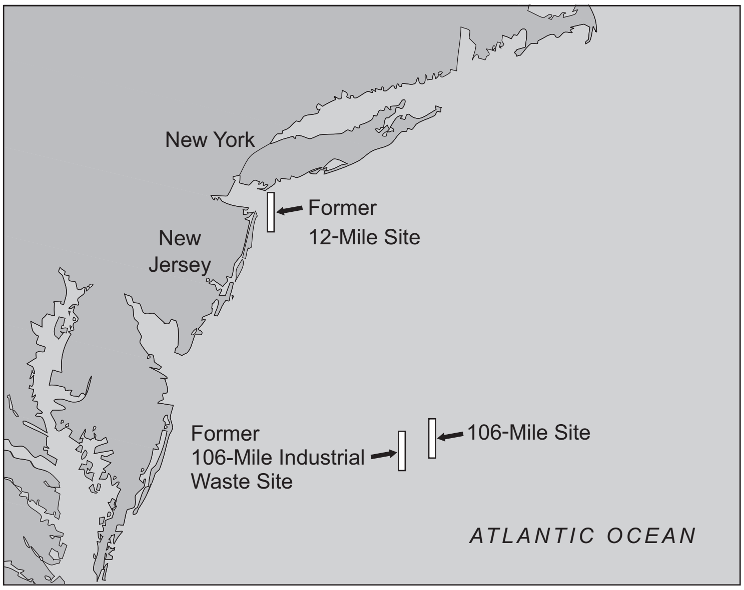
6

3430UA01 03

2. In the past, many countries, including the UK, have disposed of sewage sludge in the open ocean. A famous example of this practice is the '106 mile' dump site in the North West Atlantic. This site, 106 miles off the east coast of the USA, served the populations of New York and New Jersey. Prior to the use of the '106 mile' dump site sewage sludge was disposed of at the '12 mile' dump site.

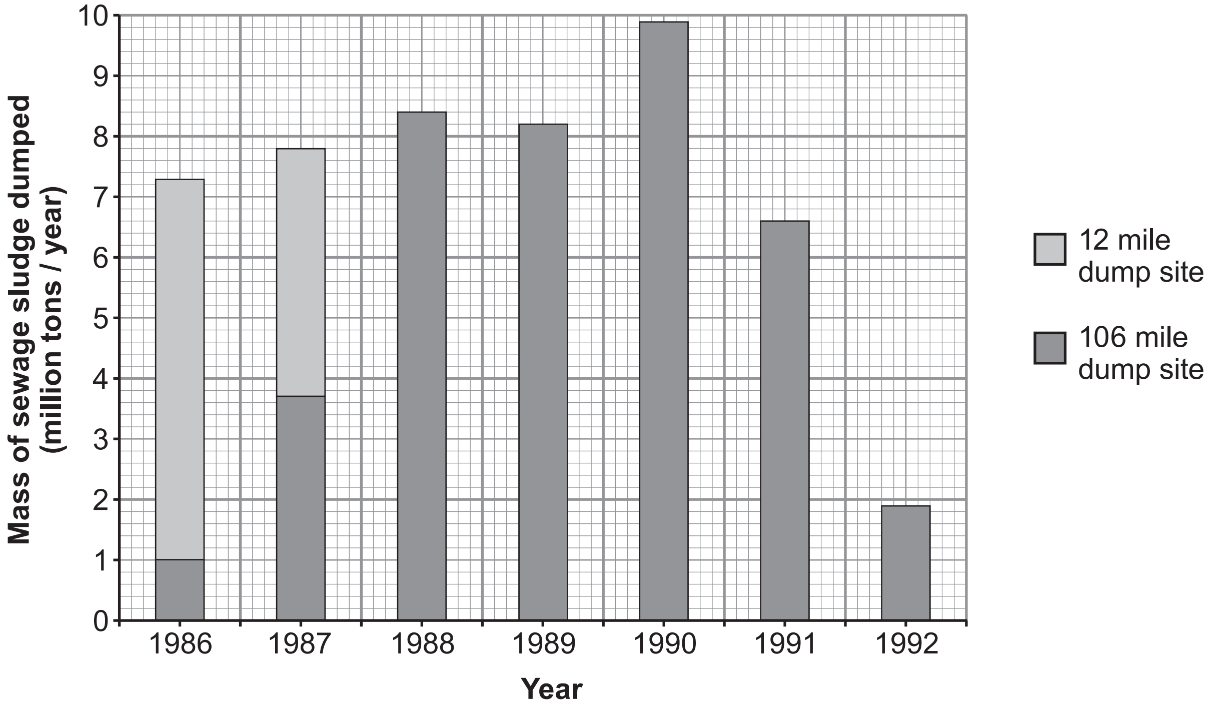
(All data from the US Environmental Protection Agency Report to Congress Sept 1995)

Map **1** showing the east coast of the USA together with disposal sites



Examiner only

Graph showing the annual disposal of sewage sludge at the ‘12 mile’ and ‘106 mile’ dump sites from 1986 to 1992.



3430UA01 05

(a) (i) Calculate the total mass of sewage sludge disposed of at the ‘12 mile’ dump site in 1986 and 1987. [2]

total mass of sewage sludge = ... million tons

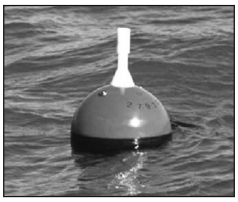
(ii) What was the final year in which sewage sludge was disposed of at the ‘12 mile’ dump site? [1]

....

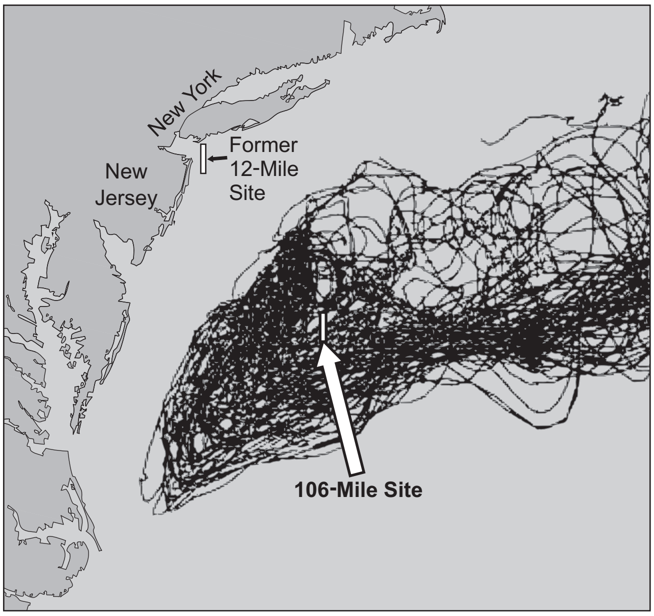
Examiner only

(iii) The US Environmental Protection Agency released buoys into the ocean at the 106 mile dump site. They used satellites to track the movement of the buoys between 1989 and 1992.

drifting buoy tracked by satellite



Map **2** showing the movement of buoys



Use the information in map **2** to suggest why it was decided to select a sewage dump site 106 miles off the east coast of the USA and to close the '12 mile' dump site. [2]

....

....

....

....

....

Examiner only

(b) Of the sewage sludge which is dumped, 20 – 70% reaches the sea bed. Here the oxygen consumed by living organisms increases greatly. Explain why this happens. [3]

....

....

....

....

....

(c) Close to the ‘106 mile’ sewage dump site is the former ‘106 mile’ **industrial** waste dump site. (See map **1** on page 4). Name **one** group of industrial wastes which you would expect to find at this site. [1]

....

9

3430UA01 07

Examiner only

3. Describe the functions of bile and lipase in the breakdown of fats. [6 QER]

6

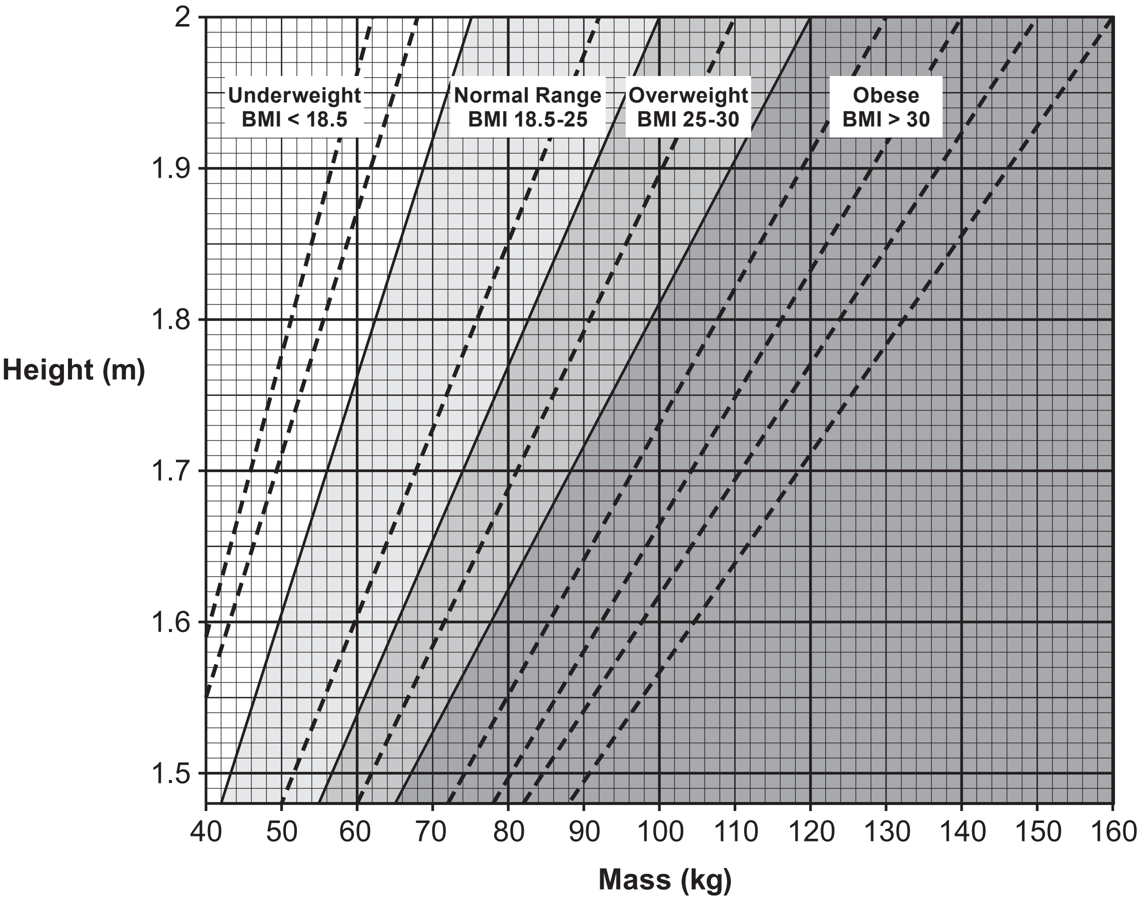
Examiner only

4. The Body Mass Index (BMI) is one way of measuring whether a person is a healthy mass for their height. The result gives an indication of whether the person is a healthy mass, and if not, how overweight or underweight they are.
The BMI is calculated from the mass and height of a person. It is calculated using the following equation:

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

where body mass is measured in kilograms and height in metres.

The chart shows the BMI categories based on the World Health Organisation data. The dashed lines represent subdivisions within a major category. For example, the underweight category is subdivided into severe, moderate and mild.



Source: Calculator.net

(a) From the chart, state how many subdivisions there are within the obese category. [1]

number of subdivisions in the obese category = ...

 (3430UA0-1)

Examiner only

(b) The table below shows some biometric data for three teachers at a school.

Teacher	Mass (kg)	Height (m)	BMI	BMI category
Iorwerth	82.4	1.56	33.9	obese
Sharon	55.1	1.71	18.8	....
Peter	95.3	1.84	....	....

(i) Calculate the BMI for Peter and **write the answer in the table**. [2]

Space for working.

3430UA01 11

(ii) Use the chart opposite to allocate a major BMI category to Sharon and Peter and **write the answer in the table** above. [1]

(c) Iorwerth wants to reduce his BMI. State **two** ways in which he could alter his diet in order to achieve this. [2]

I. ....

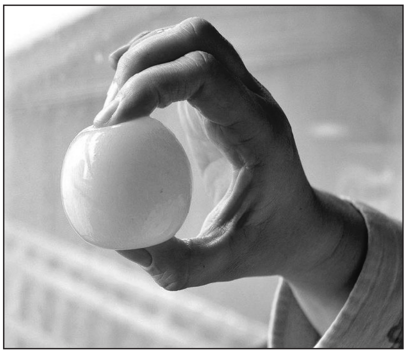
II. ....

6

Examiner only

5. A hen's egg is a single cell which is surrounded by a calcium carbonate shell.
Six hen's eggs were soaked in 4% ethanoic acid for 24 hours to dissolve their shells. After this time each egg is still intact as it is surrounded by a cell membrane. The initial masses of the 'naked eggs' (eggs without shells) were recorded.

'naked egg'



The six 'naked eggs' were each placed in sucrose (sugar) solutions of different concentrations and left for 24 hours. After this time the final masses of the 'naked eggs' were recorded.

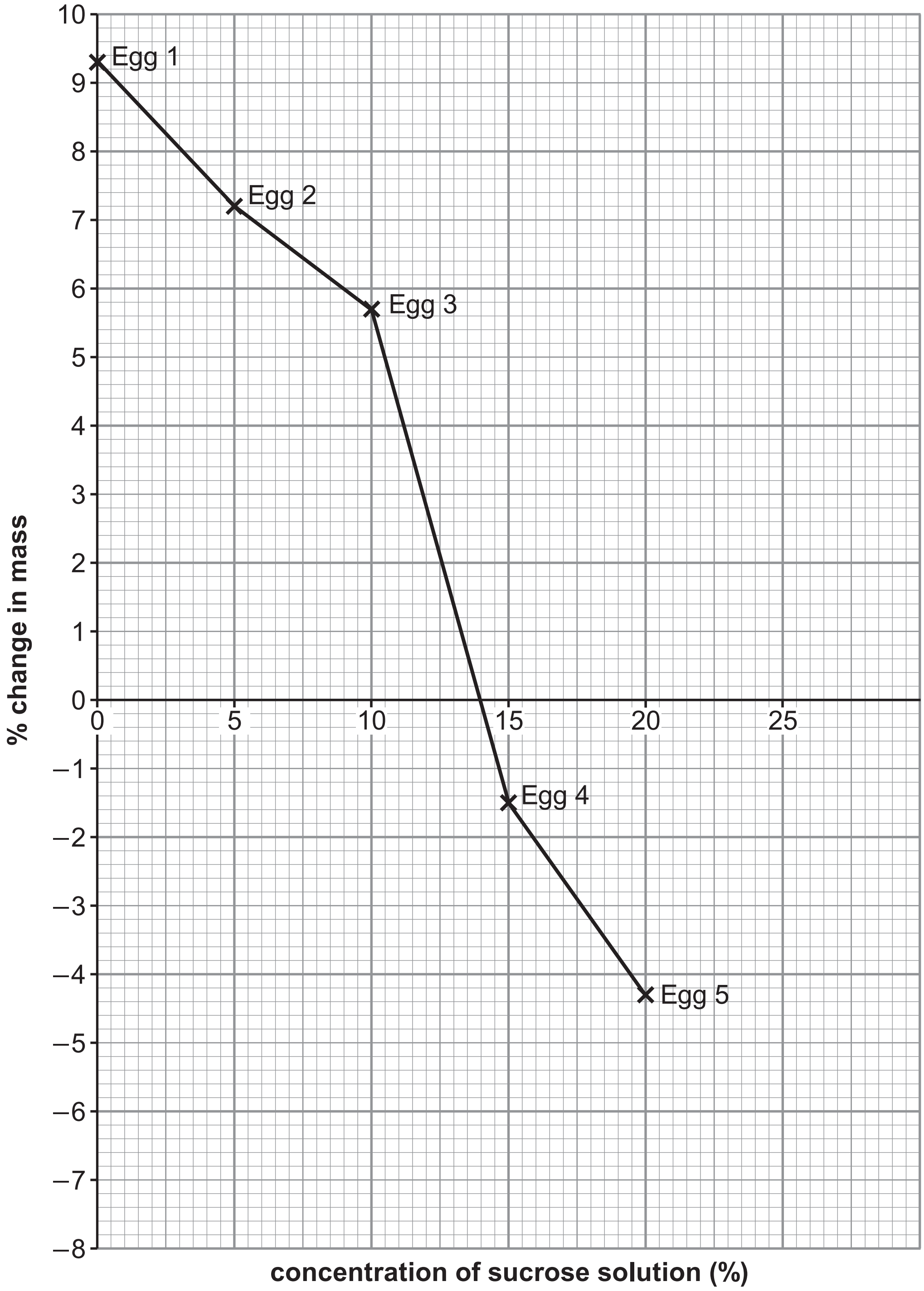
Results table

Egg number	Concentration of sucrose solution (%)	Initial mass (g)	Final mass (g)	Change in mass (g)	% change in mass
1	0	61.5	67.2	5.7	9.3
2	5	64.3	68.9	4.6	7.2
3	10	59.8	63.2	3.4	5.7
4	15	60.7	59.8	–0.9	–1.5
5	20	62.6	59.9	–2.7	–4.3
6	25	60.9	56.1	–4.8	....

(a) **Complete the table** above by calculating the % change in mass for egg number **6**. [2]
Space for working.

Examiner only

(b) In the graph below the % change in mass of the 'naked eggs' is plotted against the concentration of sucrose solution.



(i) **Complete** the **graph** by adding the percentage change in mass for egg number **6**. [1]

(ii) From the graph state the concentration of sucrose solution where there was no net movement of water. [1]

concentration of sucrose solution = ... %

Examiner only

(c) Explain the results for egg number **3** and egg number **5**. [4]

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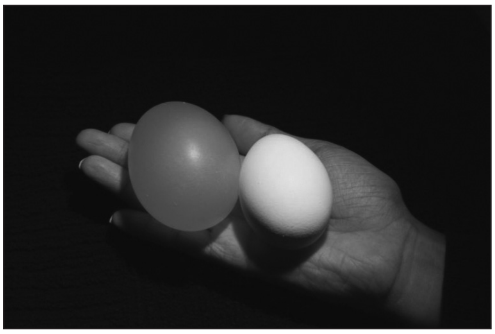
(d) The photograph below shows a complete hen's egg on the right and one of the 'naked eggs', used in this experiment, on the left.

Suggest a concentration of sucrose solution in which this 'naked egg' was kept. [1]

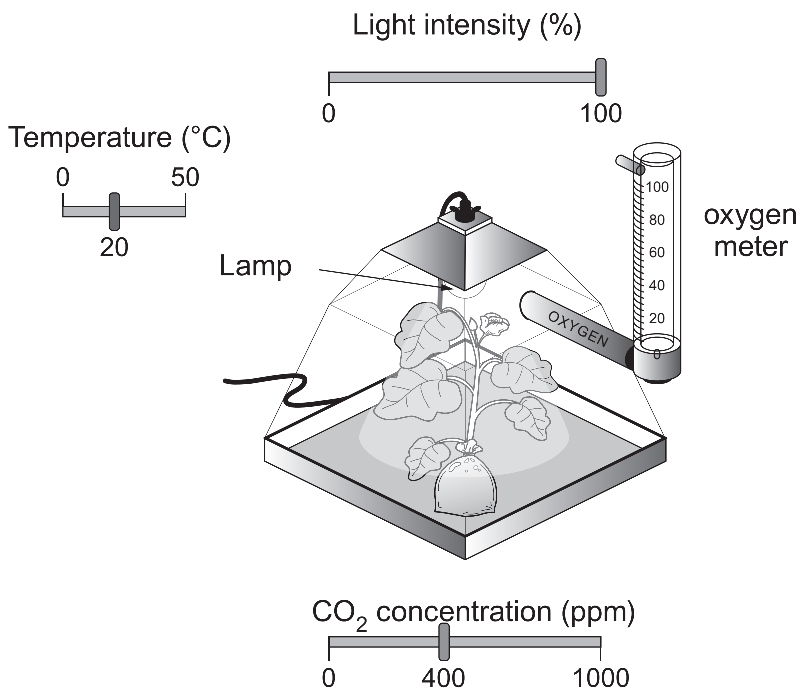


suggested concentration of sucrose solution = ... %

9

6. In a classroom exercise some students used a computer simulation to investigate the factors affecting the rate of photosynthesis. A well-watered plant, was placed inside a transparent sealed container with a high intensity lamp above the plant. The air in the airtight container was continually monitored in order to measure the rate of O_2 production

- The light intensity of the lamp could be varied between 0 - 100 %.
- The CO_2 concentration of the air inside the sealed container could be varied between 0 – 1000 ppm.
- The temperature inside the sealed container could be varied between 0 – 50 °C.



The table below shows the results obtained by one of the students.

Reading number	Light intensity (%)	Temperature (°C)	CO_2 concentration (ppm)	O_2 production (cm^3/h)
1	0	10	0	0.0
2	20	10	200	3.1
3	40	10	200	3.1
4	40	10	400	3.1
5	40	20	400	34.7
6	60	20	400	41.7
7	80	20	400	41.7
8	100	25	400	41.7
9	100	25	600	47.3

Examiner only

(a) Explain how oxygen production can be used as a measure of the rate of photosynthesis. [3]

(b) Explain why oxygen production remains at 3.1 cm^3/h for readings **2**, **3 and 4**. [2]

(c) Identify the limiting factor for readings **6**, **7**, **8 and 9** and explain your answer. [2]

(d) State why the container must be sealed. [1]

Examiner only

(e) Sian and Dafydd are students who carried out the computer simulation. Sian suggested that they try to set up a 'live version' of the apparatus in the laboratory. Dafydd said that if they did this, a problem could arise which would affect the validity of the experiment. Suggest the problem that may arise and how this could be solved. [2]

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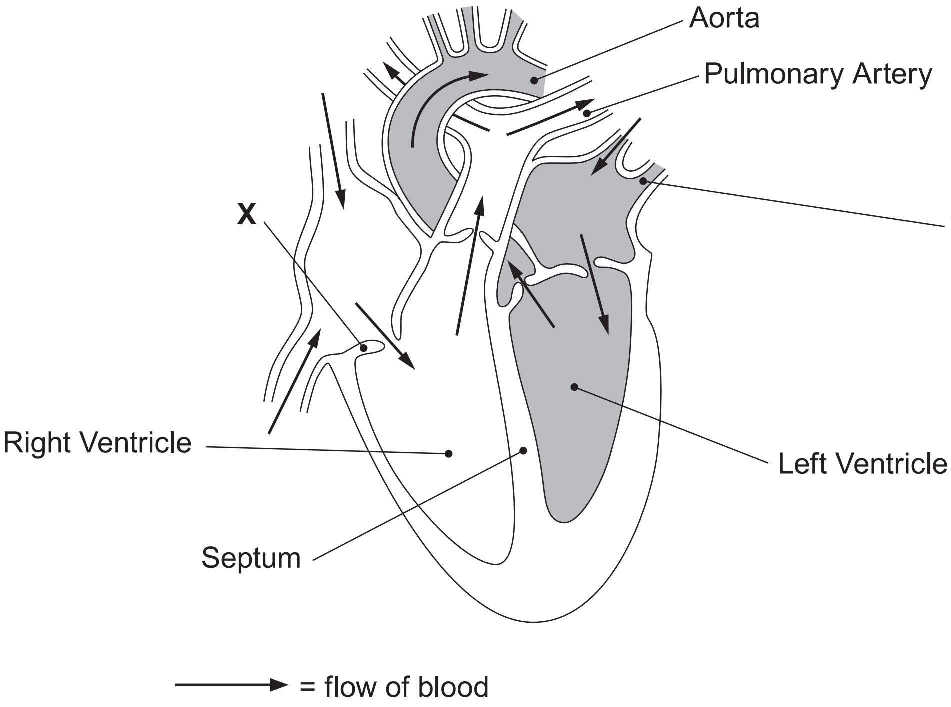
10

Examiner only

7. The diagram shows a section through the human heart.



(a) State the name of structures labelled **X** and **Y** on the diagram. [2]

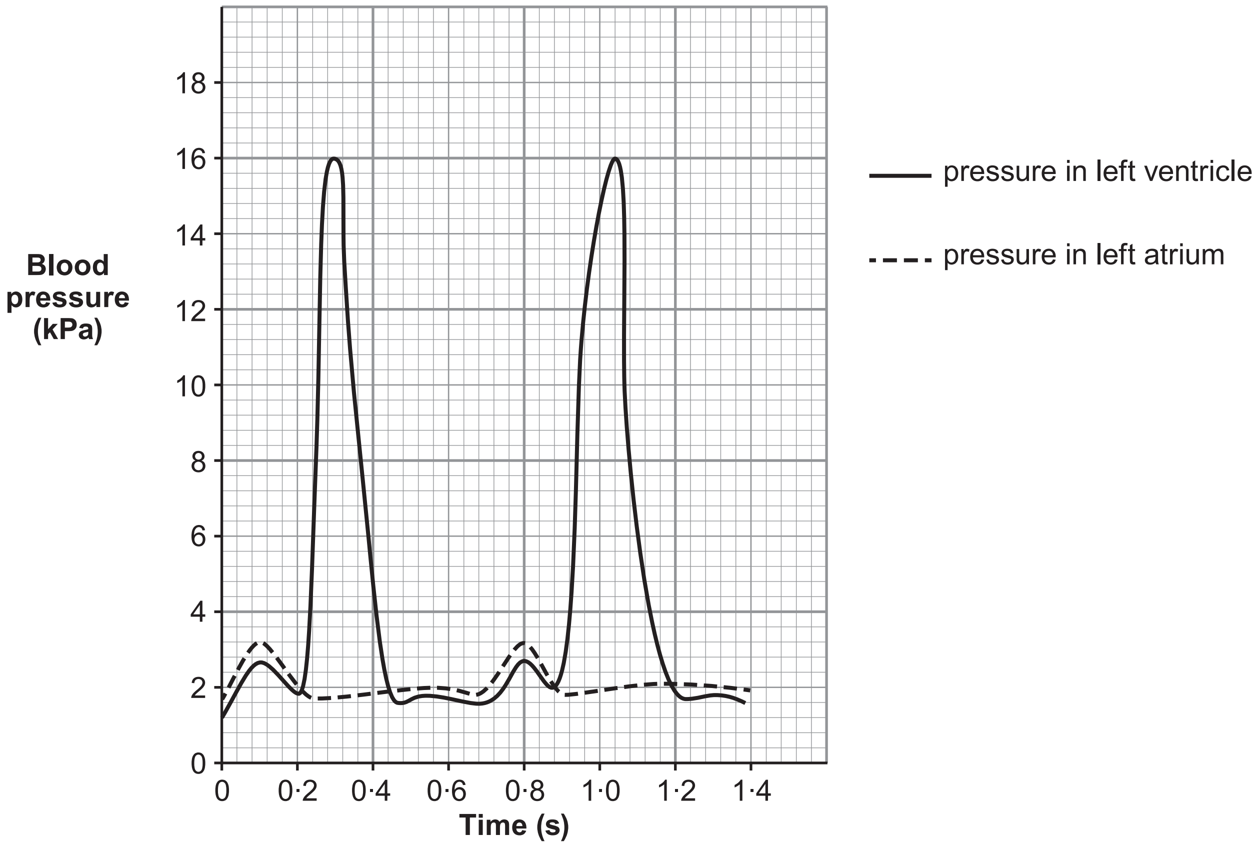
X ....

Y ....

Examiner only

(b) The walls of the atria and ventricles are of different thicknesses and are made of muscle. When the muscle contracts blood is pumped from one part of the heart to another or is pumped out of the heart.

The chart below shows the change in the blood pressure in the left atrium and left ventricle as they contract.



(i) State the maximum blood pressure in: [1]

I. left atrium .... kPa

II. left ventricle .... kPa

(ii) State how the blood pressure in the right ventricle would be different to that in the left. Give the reason for this difference and explain its significance. [3]

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6

 (3430UA0-1) **Turn over.**

Examiner only

8. The word equation for aerobic respiration is shown below.

glucose + oxygen ⟶ carbon dioxide + water + ATP

(a) In the space below state the word equation for anaerobic respiration in human cells. [1]

....

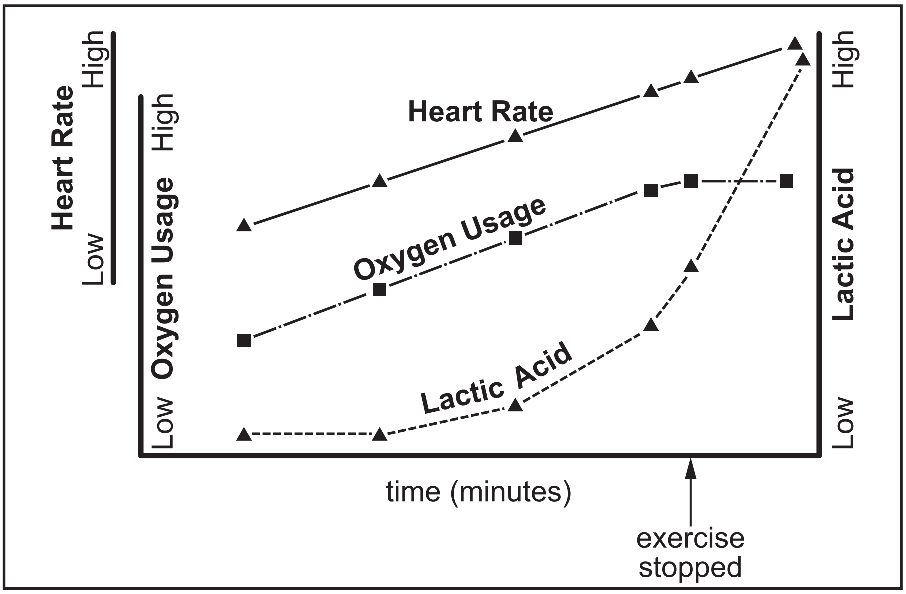
(b) Explain why anaerobic respiration is less efficient than aerobic respiration. [2]

....

....

....

(c) An Olympic athlete exercised on a treadmill. During the exercise her blood oxygen and lactic acid levels were continuously monitored as was her heart rate.
The athlete's fitness coach knew the maximum intensity of exercise she could perform (100%). The athlete increased the intensity of exercise until she reached the maximum intensity of exercise she could perform (100%). She then stopped the exercise but her heart rate and blood levels continued to be monitored.
The graph below appeared on a computer screen.



Examiner only

(i) Explain why, even after exercise stops, the athlete continues to take in and use large volumes of oxygen. [2]

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....

....

(ii) Explain why the heart rate remains high after exercise stops. [1]

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(d) The ability of oxygen to pass into the capillaries around the alveoli decreases with increasing altitude. Above 1500 m physical activity becomes more difficult.
In 1968, the Olympic Games were held in Mexico City (altitude 2268 m). Athletes and their coaches realised the difficulty of competing at the altitude of Mexico City and many of them arrived, and started training, three months before the games started. As a result of this long period of acclimatisation to altitude, the number of red blood cells per unit volume of blood in the athlete's body increased. This is one of the effects of living and working at altitude.
Explain the advantage, to the athletes, of the increased number of red blood cells per unit volume of blood. [2]

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8

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3430UA0-1

S19-3430UA0-1

FRIDAY, 7 JUNE 2019 – AFTERNOON

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**9**	
3.	**11**	
4.	**14**	
5.	**7**	
6.	**6**	
7.	**7**	
Total	**60**	

3430UA01
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

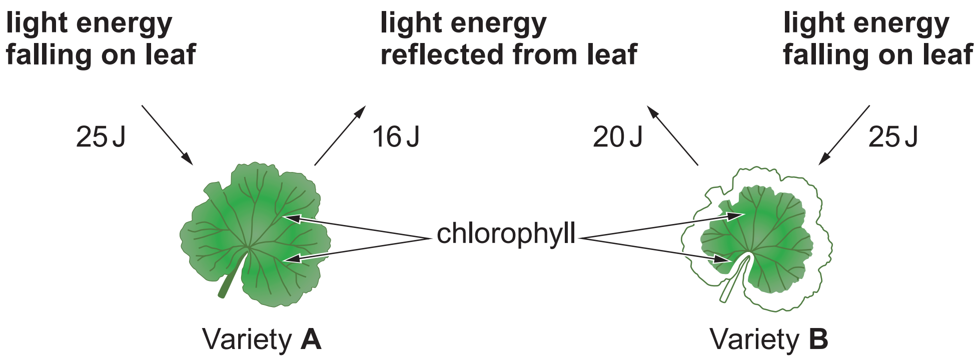
Examiner only

Answer **all** *questions.*

1. *(a)* Write the word equation for photosynthesis. [2]

....

(b) The diagram shows leaves from two varieties of geranium **A** and **B** of the genus *Pelargonium*. Light energy falling on the two leaves and the light energy reflected from each leaf, during a period of one hour is also shown.



% light energy reflected

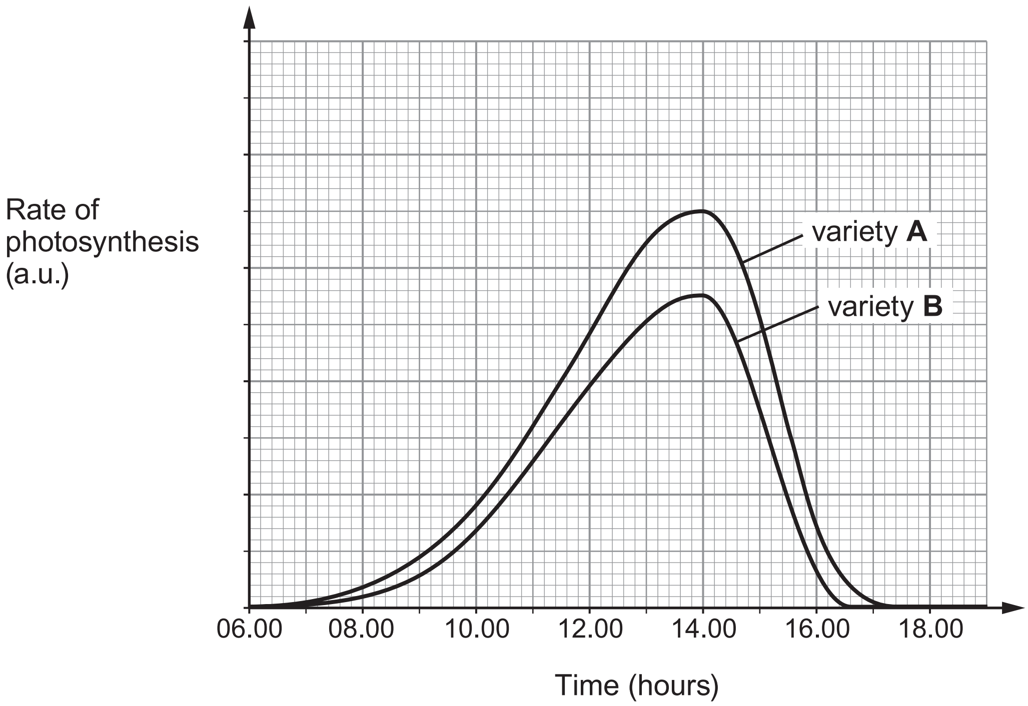
Variety **A** = 64% Variety **B** = ... %

Calculate the percentage of light energy falling on leaf **B** that is reflected.
Write your answer in the diagram. [2]

Space for working.

Examiner only

(c) The graph shows the rate of photosynthesis in the two plants between 06.00 and 18.00.



Use the diagram opposite and the graph above to explain why the mass of variety **A** is likely to increase at a faster rate than the mass of variety **B**. [2]

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Examiner only

2. *(a)* (i) State the products of protein digestion. [1]

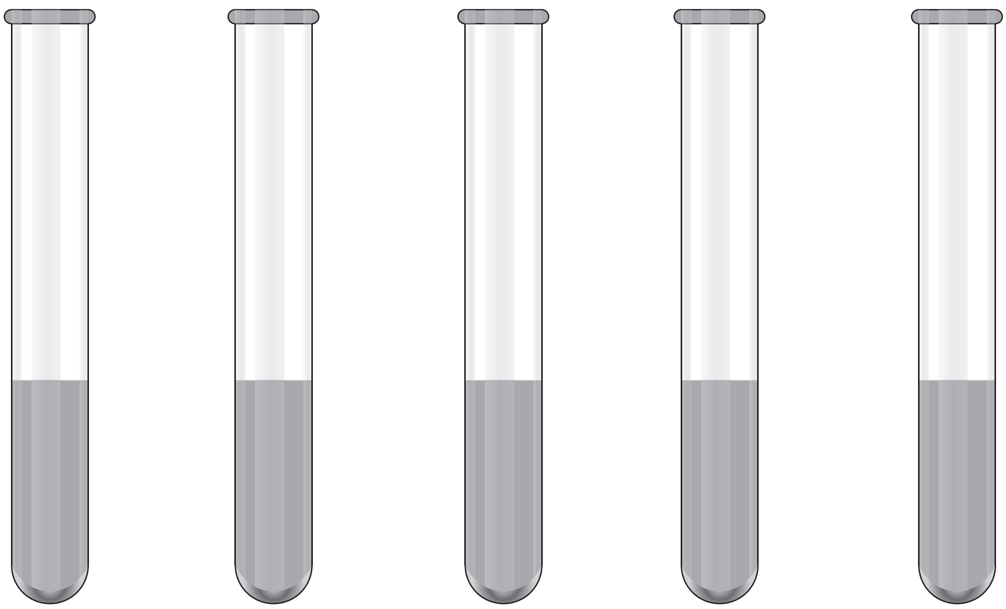
....

(ii) State the part of the digestive system that absorbs digested food molecules. [1]

....

(b) Protein digestion starts in the stomach.

Students investigated protein digestion. They set up five tubes **A** to **E**, as shown below.



Tube **A** **B** **C** **D** **E**

<table>
<tr><th>Tube A</th><th>Tube B</th><th>Tube C</th><th>Tube D</th><th>Tube E</th></tr>
<tr><td>5 cm³ of 1% protein</td><td>5 cm³ of 1% protein</td><td>5 cm³ of 1% protein</td><td>5 cm³ of 1% protein</td><td>5 cm³ of 1% protein</td></tr>
<tr><td>5 cm³ of 0.1% protease</td><td rowspan="3">15 cm³ liquid from the stomach</td><td>5 cm³ of 0.1% protease</td><td rowspan="2">15 cm³ distilled water</td><td rowspan="2">15 cm³ distilled water</td></tr>
<tr><td>10 cm³ distilled water</td><td>10 cm³ distilled water</td></tr>
<tr><td>pH 2.0</td><td>pH 7.0</td><td>pH 2.0</td><td>pH 7.0</td></tr>
</table>

Examiner only

The students measured the percentage protein digested at one hour. The results are shown in the table.

Tube	Percentage protein digested
A	100
B	98
C	5
D	0
E	0

(i) Compare the results for tubes **A** and **D**. State the conclusion that can be made about the digestion of protein. [1]

....

....

(ii) State the conclusions that can be made about the contents of the liquid from the stomach in Tube **B**. [2]

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(iii) All the 1% protein added to Tube **A** had been fully digested after one hour. State why the contents of Tube **A** would still test positive for protein. [1]

....

....

(iv) The students carried out a similar investigation, but instead of using 0.1% protease, they used 5 cm^3 of 0.1% lipase in both tubes **A** and **C**.
They found there had been no digestion of protein in either tube.
Use your knowledge of enzyme structure and function to explain the results that the students obtained. [2]

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(v) State **one** further variable that should be controlled during this investigation. [1]

....

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Examiner only

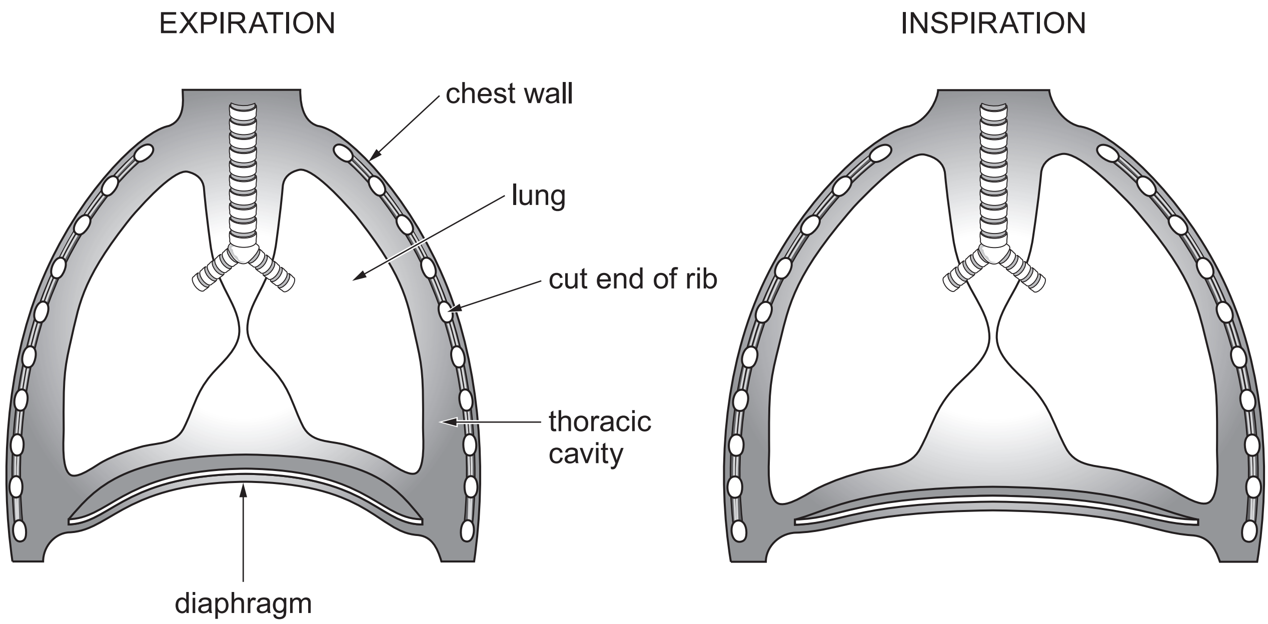
3. The thoracic cavity is the space in the body that contains the lungs. The volume of this space in a healthy adult man is about 5 500 cm^3. The lungs occupy 70% of the thoracic cavity. This is around 3 850 cm^3.

(a) State **two** other structures which occupy the remaining 30% of the space of the thoracic cavity. [2]

I. ....

II. ....

(b) The diagrams below show front views of the thorax during expiration and inspiration.



Use the information in the diagrams above, and your own knowledge, to explain how **inspiration** occurs. [5]

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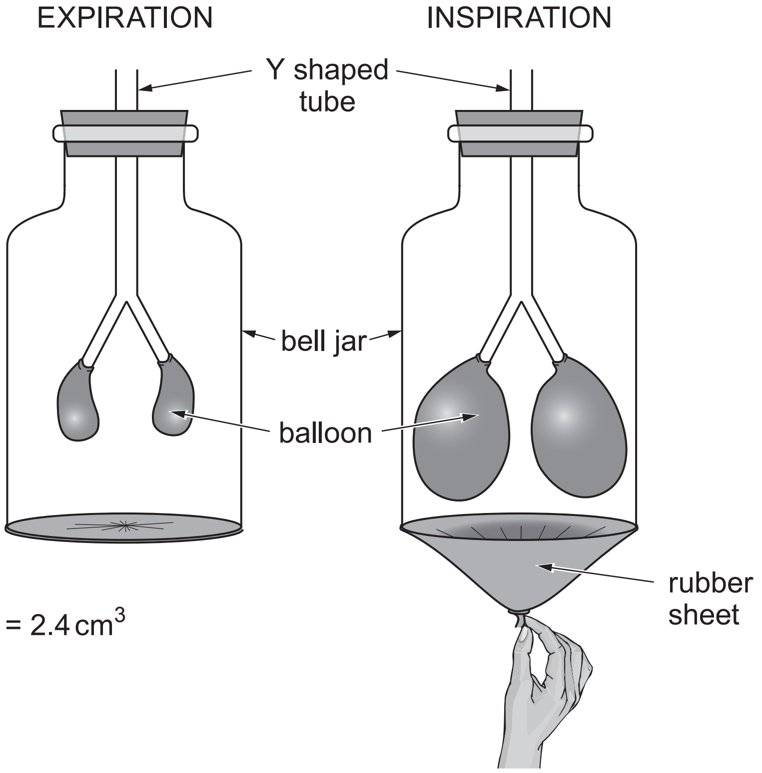
Examiner only

(c) The bell jar model can be used to demonstrate expiration and inspiration as shown below.



Volume of one deflated balloon = 2.4 cm^3

Volume of bell jar = 5 500 cm^3

(i) Calculate the percentage of the volume of the bell jar taken up by the **two** deflated balloons. Give your answer to **two decimal places**. [2]

% of volume of bell jar taken up by deflated balloons = ...

(ii) Using **only** the information and diagrams given and the calculation above, describe **two** limitations of the bell jar model compared with the human thorax. [2]

I. ....

....

II. ....

....

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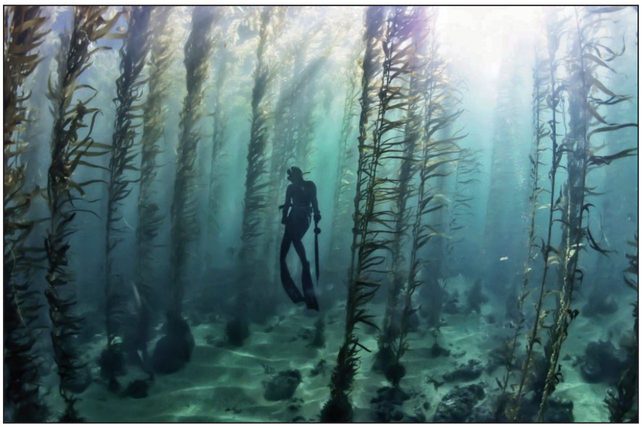
Examiner only

4. Seaweeds (marine algae) contain large amounts of iodine in their tissues. The iodine is obtained from the seawater in which they live and is needed to maintain healthy growth of the plant. The iodine content of seawater is about 0.055 mg / kg. Some of the largest seaweeds are the kelps that live in kelp 'forests' in shallow seas and oceans where light can penetrate. Some kelp species are over 80 m in length and can grow 0.5 m in a day.

Photo showing part of a kelp 'forest' with sub-aqua diver



(a) State the reason why kelp only grows in seawater where sunlight can penetrate. [1]

....

(b) Kelp is of commercial importance. It is harvested for many reasons including as a source of iodine and other chemicals for the pharmaceutical industry. Conservationists are concerned that in some parts of the world kelp harvesting is reducing the number of other organisms living in the kelp forests.

Harvesting kelp in Canada

 (3430UA0-1)

Examiner only

Using **only the information given**, what could you, as a scientist, say to the conservationists that would help them understand that a certain level of kelp harvesting could be maintained? [1]

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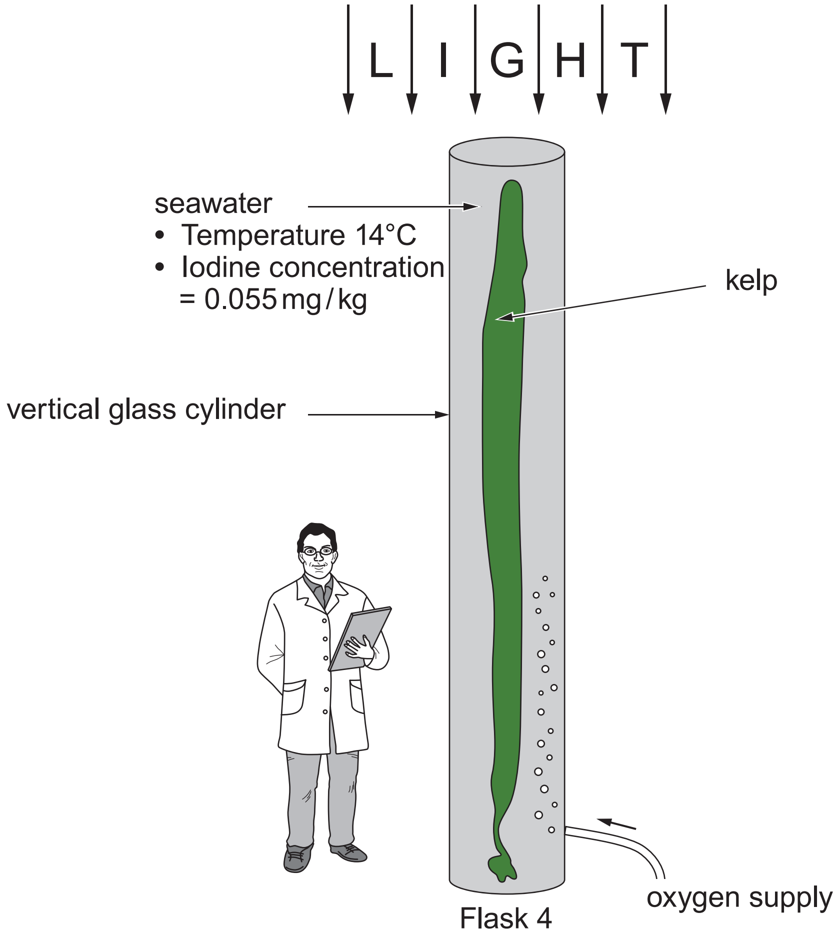
09

(c) The increasing importance of the commercial use of kelp has led scientists to conduct experiments in an attempt to grow them in land based factories. In one experiment sugar kelp *(Laminaria saccharina)* was grown in seawater in large vertical glass flasks. Scientists were trying to establish if the concentration of oxygen contained in the seawater affected the absorption of iodine by the kelp.

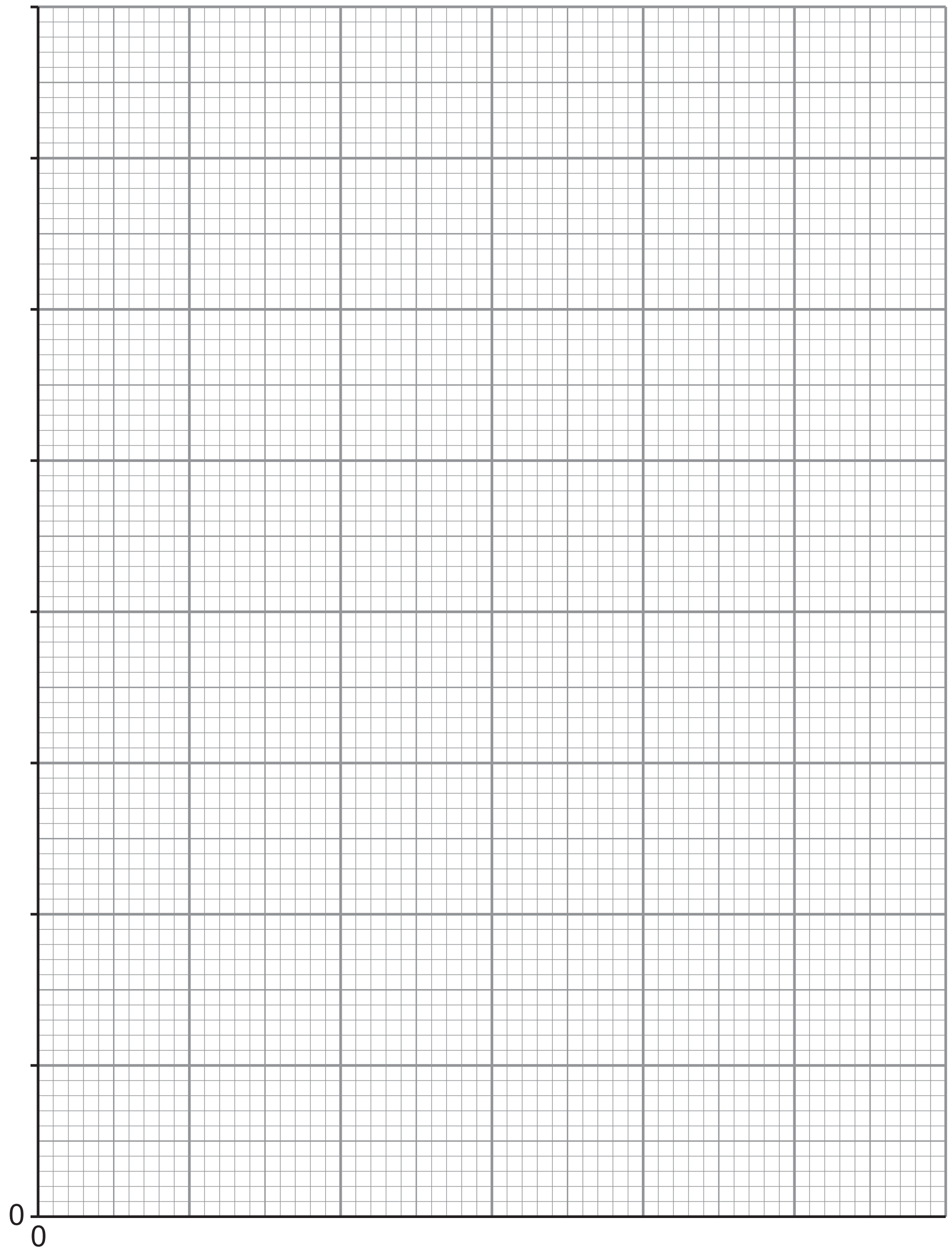


One set of results from the experiment is shown in the table.

Flask No.	Rate of flow of oxygen into flask (dm^3/minute)	Mass of iodine extracted from kelp (mg/kg)
1	0.0	45
2	0.5	140
3	1.0	259
4	1.5	676
5	2.0	740
6	2.5	780
7	3.0	780

Examiner only

(i) On the grid below plot a line graph for the mass of iodine against the rate of oxygen flow. You must add suitable scales to each axis. Join the plots with a ruler. [4]



Examiner only

(ii) Describe the effect of increasing the rate of flow of oxygen above 2.5 dm^3/minute on the mass of iodine extracted. [1]

....

....

(iii) The kelp used in the 7 flasks were of different sizes but the experimental results could still be compared. Explain how this was possible. [1]

....

....

(iv) For **Flask 4** calculate how many times greater the concentration of iodine is in the kelp compared to the concentration of iodine in seawater. **Give your answer in standard form.** [2]

Answer = ... × greater

(v) The only source of iodine for kelp is the seawater in which they live. Explain how kelp is able to accumulate iodine against a concentration gradient. [3]

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(d) Another group of scientists want to test the reproducibility of the above experiment. State **one** *other* controlled variable, the value of which they would need to know, before they could proceed. [1]

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14

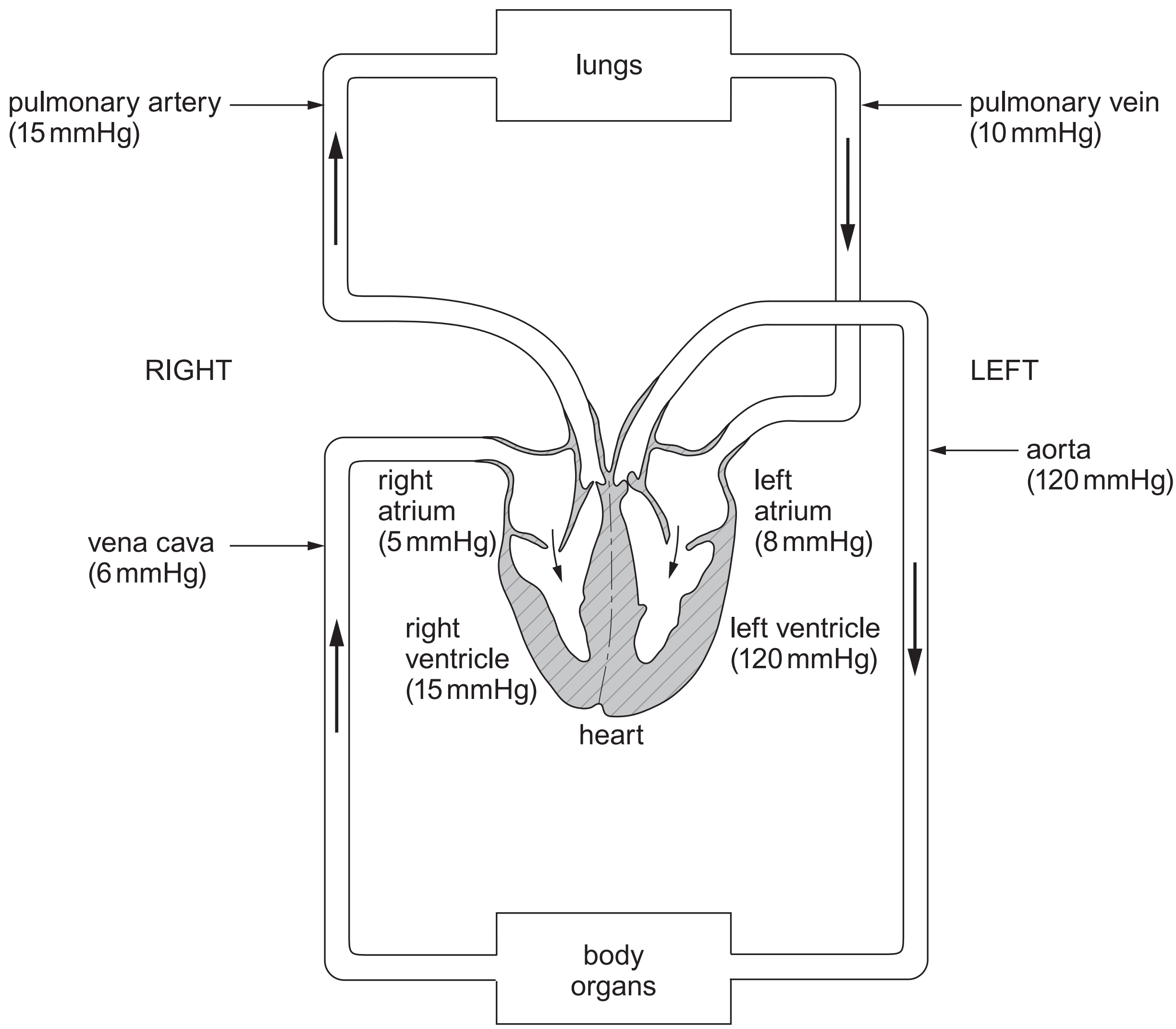
5. In one complete circulation of the body, blood passes through the heart twice. The blood travels:

- from the heart to the lungs and then back to the heart in the pulmonary circulation and then;
- from the heart to the other organs of the body and back to the heart in the systemic circulation.

This double circulation is essential for the functioning of highly active animals such as mammals.

The diagram below represents the double circulation in humans. It shows the direction of blood flow and the pressure of the blood in various blood vessels in a healthy 25 year-old woman at rest.

The pressure of the blood is measured in mmHg (millimetres of mercury).



Examiner only

(a) Suggest a reason for the difference in the:

(i) thickness of the walls of the atria and the ventricles; [1]

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(ii) blood pressure between the pulmonary artery and pulmonary vein; [1]

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(iii) blood pressure between the pulmonary artery and the aorta. [2]

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(b) Using the data for blood pressure in the **diagram opposite**, suggest why, after leaving the lungs, the blood has to return to the heart before it is sent to the other organs of the body. [2]

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(c) Describe how the direction of blood flow is maintained in the circulatory system. [1]

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7

Examiner only

6. Explain the function of cilia and mucus in the cleaning mechanism of the lungs and describe the effect smoking has upon this cleaning mechanism. [6 QER]

6

7. Carwyn and Anwen are year 10 students who are concerned that they are overweight. They think that their health may be affected in later life. They both agree to help each other follow a balanced diet.

They research the topic on the Internet and find the following information:

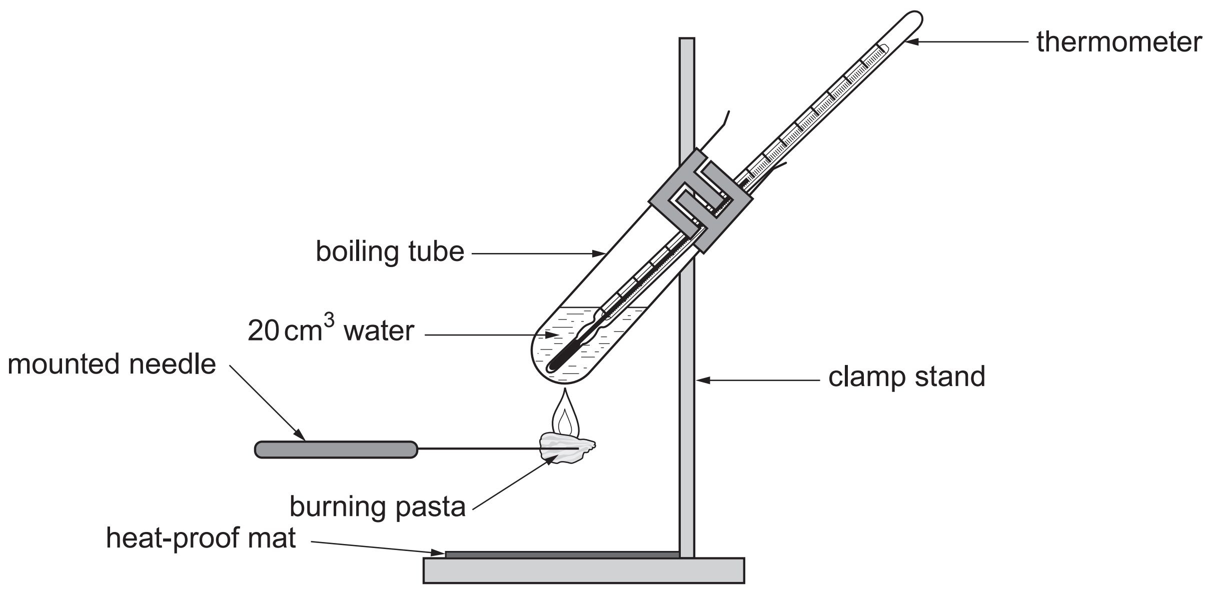
As part of a healthy balanced diet, it is recommended that people of their age should be eating the following every day:

Energy	8400 kJ
Total fat	70 g
Starchy carbohydrate (e.g. bread, pasta, rice)	260 g
Total sugar (from milk, fruit and added sugar)	90 g
Protein	50 g
Salt	6 g

Carwyn does not want more than one third of his recommended energy input per day to come from starchy carbohydrates.

For his evening meal he has decided to have pasta, the only carbohydrate he eats in the day. He wants to know the energy content of 1 g of pasta in order to calculate how much pasta to allow himself in his evening meal.

Carwyn and Anwen used the following simple calorimeter to measure the energy content of a 7 g piece of dried pasta.



Examiner only

The table shows one set of results they obtained.

Type of food	Mass of food (g)	Initial temp of water (°C)	Final temp of water (°C)	Increase in temp (°C)	Energy released per gram of food (J)	Energy released per gram of food (kJ)
Pasta	7	19	65	46	....	....

(a) (i) **Complete the table** above by using the following formula. [2]

$$\text{Energy released from pasta per gram (J)} = \frac{\text{volume of water (cm}^3\text{)} \times \text{temp increase (°C)} \times 4.2}{\text{mass of pasta sample (g)}}$$

Space for working

(ii) Calculate the mass of pasta Carwyn can eat in his evening meal to get one third of his daily energy intake. [2]

Mass of pasta = ... g

(iii) Carwyn was surprised to discover that he could have that much pasta for his evening meal. Anwen said 'there's something wrong, that's over ten bags of pasta! We have to evaluate our scientific method to find where the error lies.'

Evaluate the design of the apparatus and identify **one** source of error. [1]

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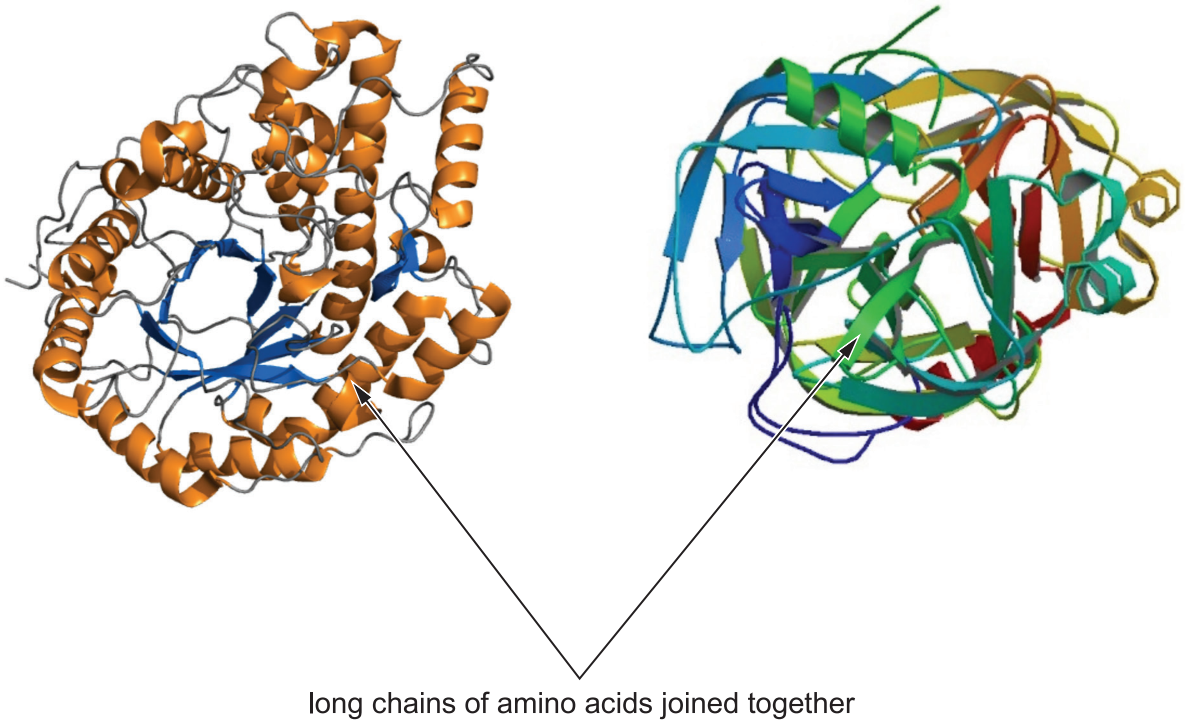
Examiner only

(b) Carbohydrase is the enzyme that digests the starch in pasta into simple sugars. Protease is an enzyme that digests protein.

Structure of a molecule of carbohydrase | Structure of a molecule of protease



With reference to the diagrams suggest how each of the enzyme molecules are different and state why it is important that the chains of amino acids are folded. [2]

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7

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3430UA0-1

Z22-3430UA0-1

WEDNESDAY, 15 JUNE 2022 – MORNING

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**6**	
3.	**8**	
4.	**12**	
5.	**6**	
6.	**12**	
7.	**7**	
Total	**60**	

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01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** questions.

1. The nutrition information in **Table 1.1** is taken from a pack of dried pasta.

Table 1.1

TYPICAL VALUES	Per 100 g (of dried pasta)
Energy	761 kJ
Fat	1.3 g
of which – saturated	0.1 g
of which – unsaturated	.... g
Carbohydrates	25.0 g
of which – sugars	2.3 g
Fibre	7.7 g
Protein	13.0 g
Salt	0.05 g

Examiner only

(a) (i) Calculate the value for unsaturated fats. **Write your answer in Table 1.1**. [1]

Space for working.

(ii) State the name of the nutrient which makes up most of the carbohydrates in the dried pasta. [1]

....

(iii) State the importance of a low-salt diet. [1]

....

....

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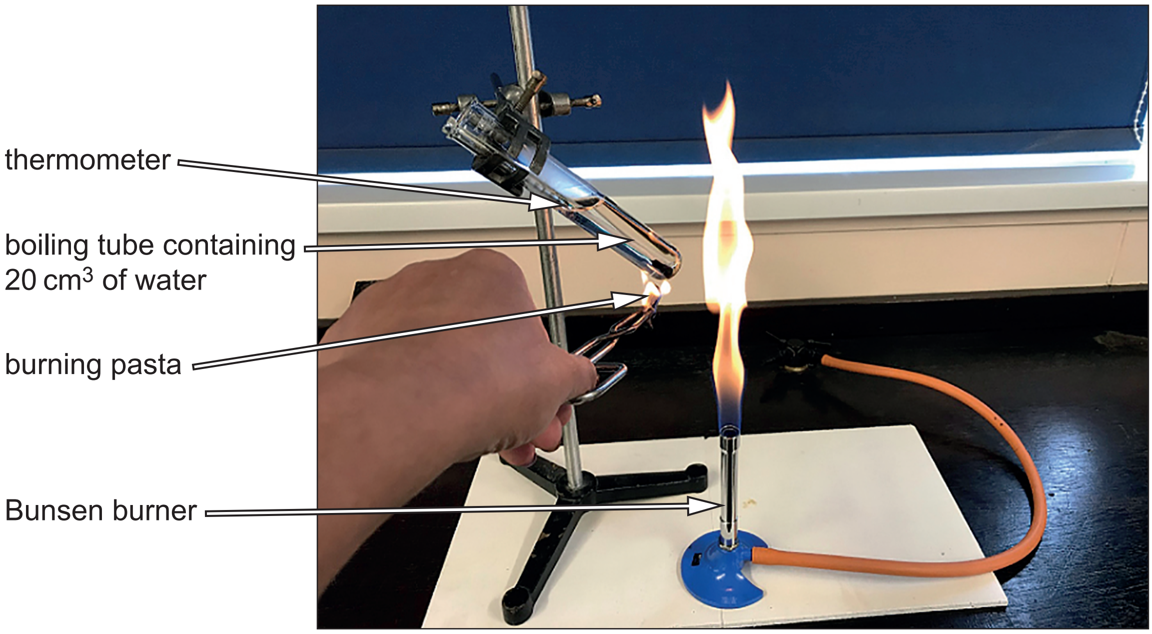
Examiner only

(b) Lloyd and Emma carried out an experiment to compare the energy values in **Table 1.1** with values they obtained using the apparatus shown in **Image 1.2**.

Image 1.2



They ignited a 1.6 g piece of dried pasta using the Bunsen burner and immediately held the burning pasta at the base of the boiling tube until it stopped burning. The results Lloyd and Emma obtained are shown in **Table 1.3**.

Table 1.3

Mass of pasta (g)	Initial temperature of water (°C)	Final temperature of water (°C)	Increase in temperature of water (°C)	Energy released per gram of food (kJ)
1.6	14	58	44	....

(i) Use the following formula to calculate the energy released per gram of food (kJ). **Write your answer in Table 1.3**. [2]

$$\text{Energy released per gram (kJ)} = \frac{\text{volume of water (cm}^3) \times \text{temperature increase (°C)} \times 0.0042}{\text{mass of pasta sample (g)}}$$

Space for working

Examiner only

(ii) I. State how the energy content of dried pasta in **Table 1.3** compares to the energy content indicated in **Table 1.1**. You must use numerical data in your answer. [2]

....

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....

II. Give **one** reason for the difference between the energy content of dried pasta obtained by Lloyd and Emma, as shown in **Table 1.3** and the energy content indicated in **Table 1.1**. [1]

....

....

(iii) Evaluate the arrangement of the apparatus shown in **Image 1.2** by identifying **one** source of error. [1]

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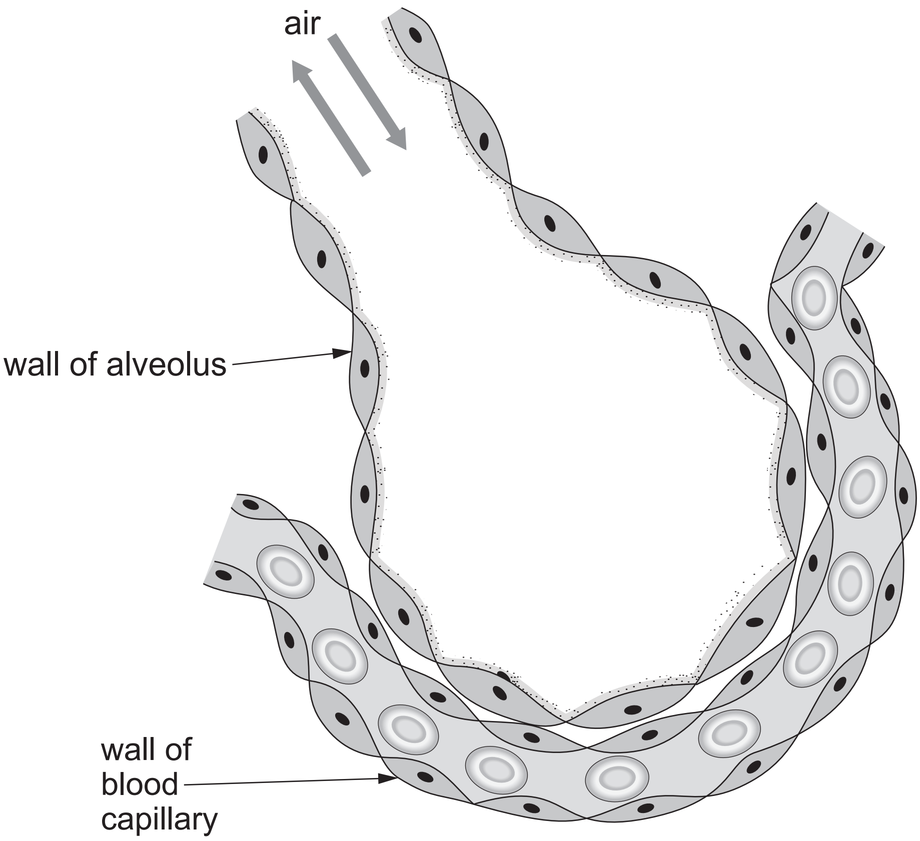
Examiner only

2. **Image 2.1** shows an alveolus.

Image 2.1



(a) **Use labelled arrows** to identify the following structures on **Image 2.1**: [2]

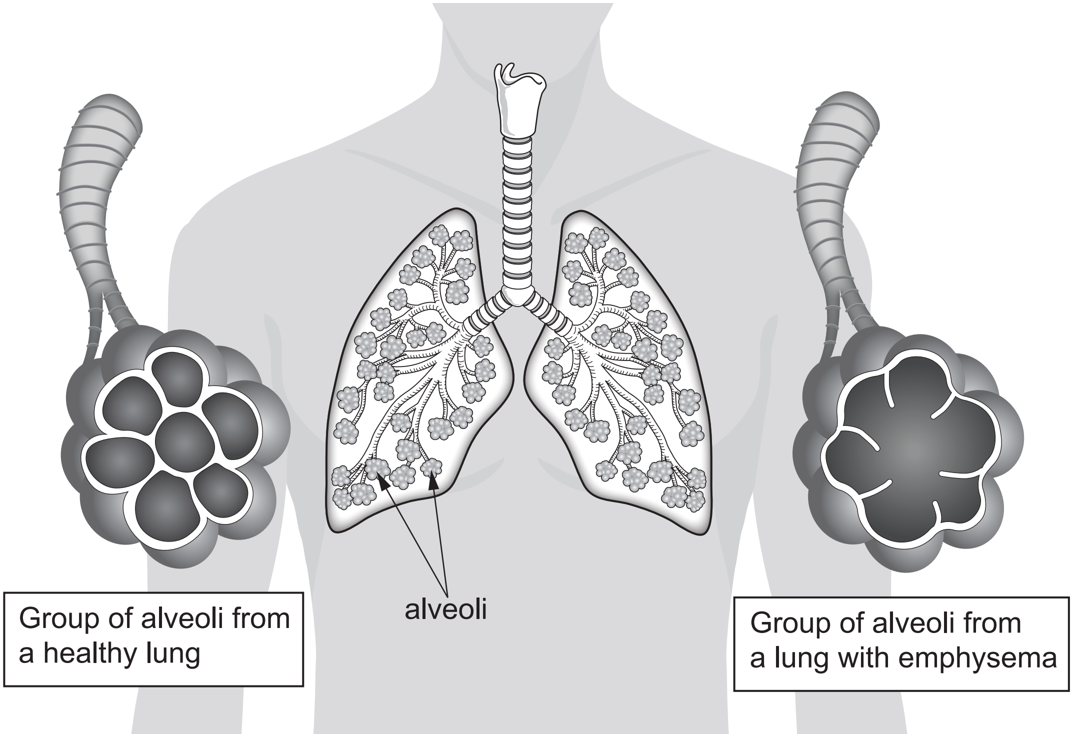
(i) bronchiole;

(ii) blood plasma.

3430UA01 07

(b) **Image 2.2** shows groups of alveoli from a healthy lung and from a lung of a person with emphysema. The tables below **Image 2.2** show the concentrations of oxygen and carbon dioxide in the blood capillaries.

Image 2.2



Healthy lung	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	86
carbon dioxide	45

Lung with emphysema	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	53
carbon dioxide	62

Examiner only

(i) Using **Image 2.2**, explain the differences in the concentrations of gases in the blood capillaries of a healthy lung and a lung with emphysema. [2]

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....

(ii) State the effect on breathing of the difference in concentrations of these gases for a person suffering from emphysema. [1]

....

(iii) State **one** cause of emphysema. [1]

....

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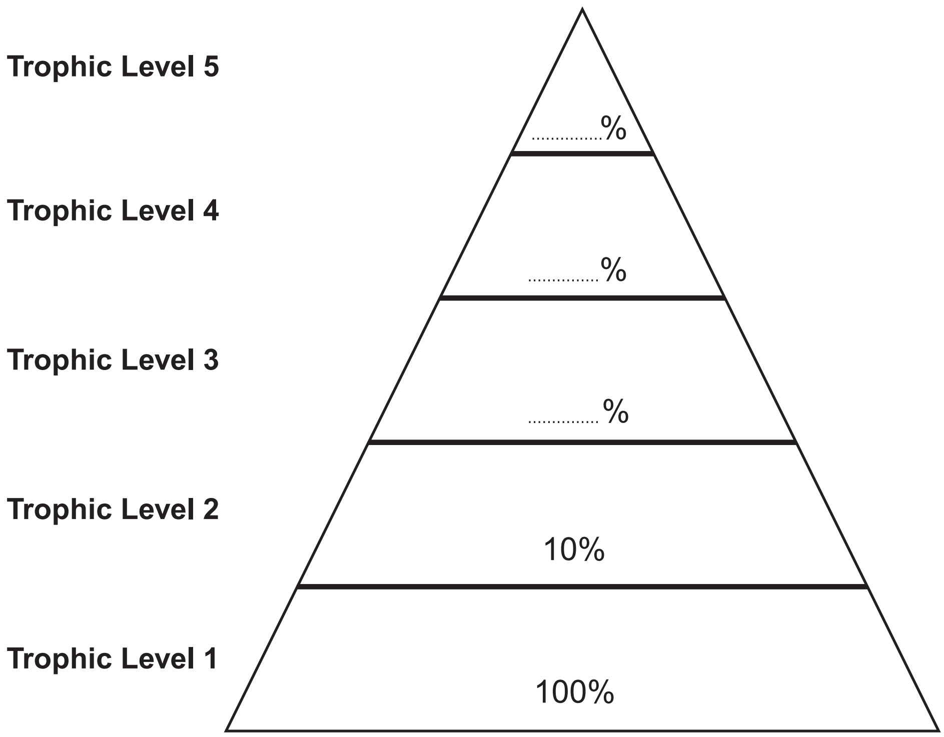
6

Examiner only

3. **Image 3.1** represents a five-stage food chain drawn as a pyramid of biomass. The energy retained or stored in the form of new biomass in trophic levels **1** and **2** is represented as a percentage.

Image 3.1



(a) State:

(i) the source of energy for this food chain. [1]

....

(ii) the scientific term used to describe the organisms found at trophic level **1**. [1]

....

(iii) a term that can be used to describe the organisms at trophic levels **2–5**. [1]

....

Examiner only

(b) (i) The efficiency of energy transfer between one trophic level and the next is **10%**. Calculate the percentage of the energy entering trophic level **1** that reaches trophic levels **3**, **4** and **5**. **Write your answers in the pyramid in Image 3.1**. [2]

Space for working.

(ii) Suggest why the food chain represented by the pyramid of biomass in **Image 3.1** could not sustain a trophic level **6**. [1]

....

....

(iii) State **one** way in which energy is lost from a food chain. [1]

....

(c) <u>Underline</u> **one** correct statement about pyramids of biomass from the following list. [1]

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As you go up most pyramids of biomass from one trophic level to the next the:

- size of the organisms decreases and their numbers increase.
- size of the organisms decreases and their numbers decrease.
- size of the organisms increases and their numbers increase.
- size of the organisms increases and their numbers decrease.

8

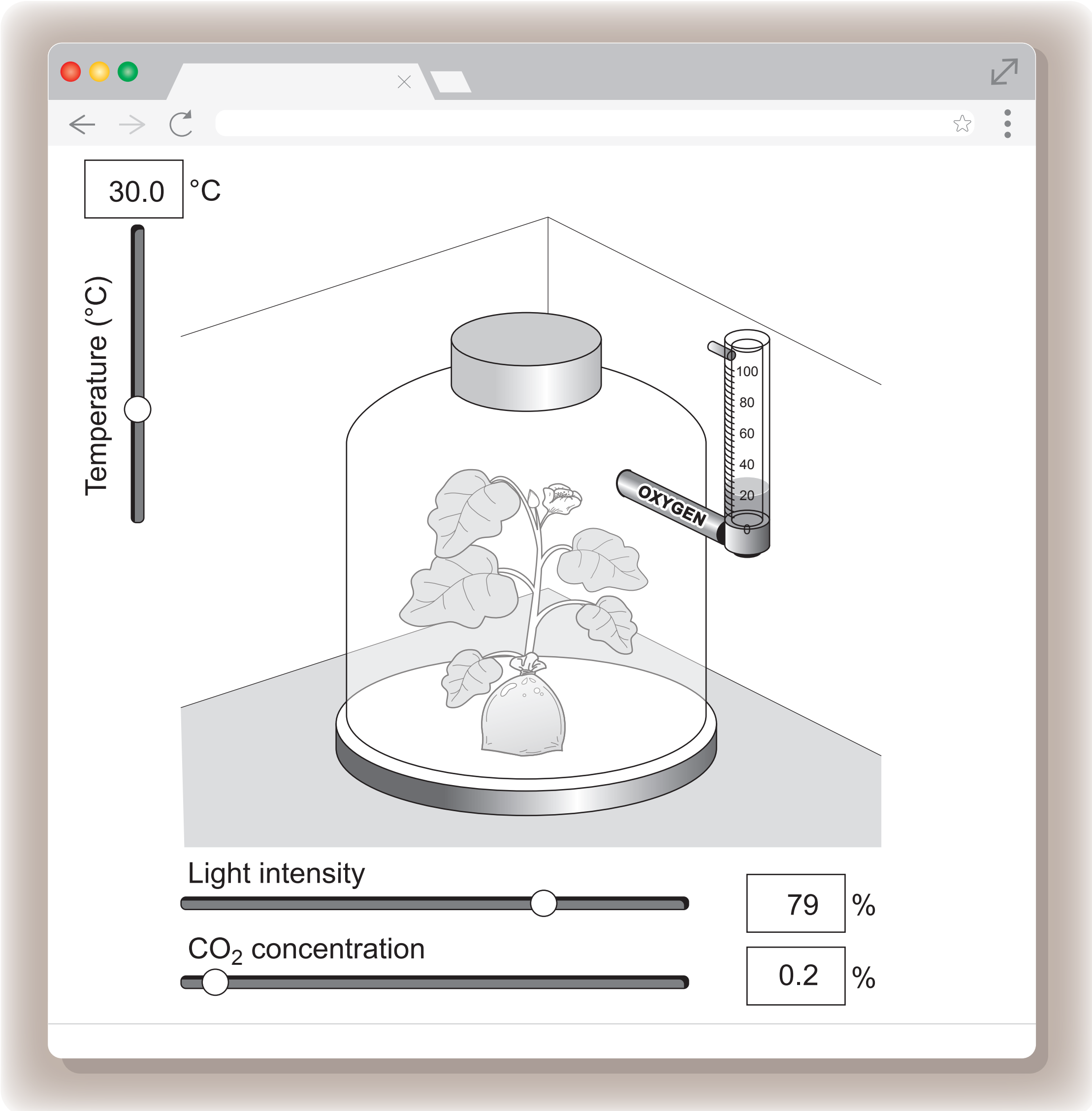
Examiner only

4. (a) State the word equation for photosynthesis. [2]

....

(b) Students used a computer simulation to investigate how the limiting factors of photosynthesis affect its rate. A screenshot from the simulation is shown in **Image 4.1.**

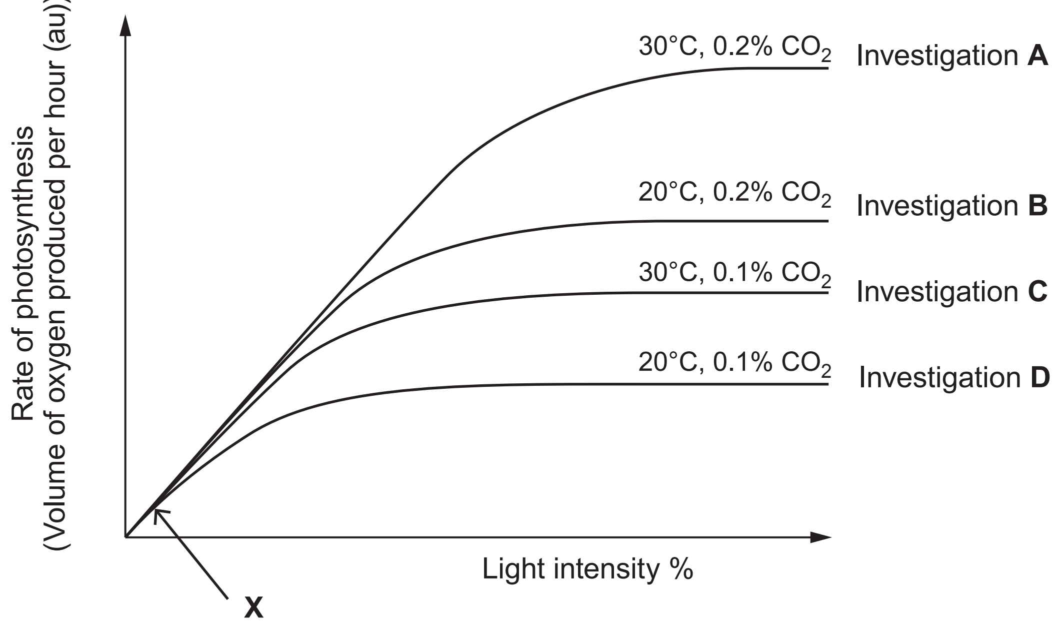
Image 4.1



The simulation allowed the students to carry out investigations into the rate of photosynthesis by adjusting the temperature, light intensity and carbon dioxide concentration the plant received. The rate of photosynthesis was then calculated by measuring the volume of **oxygen** produced by the plant per hour. The computer then produced a graph of the results as shown in **Graph 4.2**.

Examiner only

Graph 4.2



(i) With reference to limiting factors, explain why the rate of photosynthesis is:

I. lower in investigation **C** than in investigation **A**; [2]

....

....

....

....

II. higher in investigation **C** than in investigation **D**. [2]

....

....

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(ii) Suggest how you could increase the rate of photosynthesis in investigation **A**. [1]

....

....

(iii) State the limiting factor which is controlling the rate of photosynthesis at point **X** on **Graph 4.2**. [1]

....

Examiner only

(iv) The levels were adjusted to the following:

- light intensity to bright daylight levels
- CO_2 concentration to 0.04%
- temperature to 80 °C.

I. Suggest how a temperature of 80 °C would affect the rate of photosynthesis and explain your answer. [2]

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II. Suggest why a CO_2 concentration of 0.04% was chosen. [1]

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(c) Explain why the volume of oxygen produced by the plant per hour can be used as a measurement of the rate of photosynthesis. [1]

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12

Examiner only

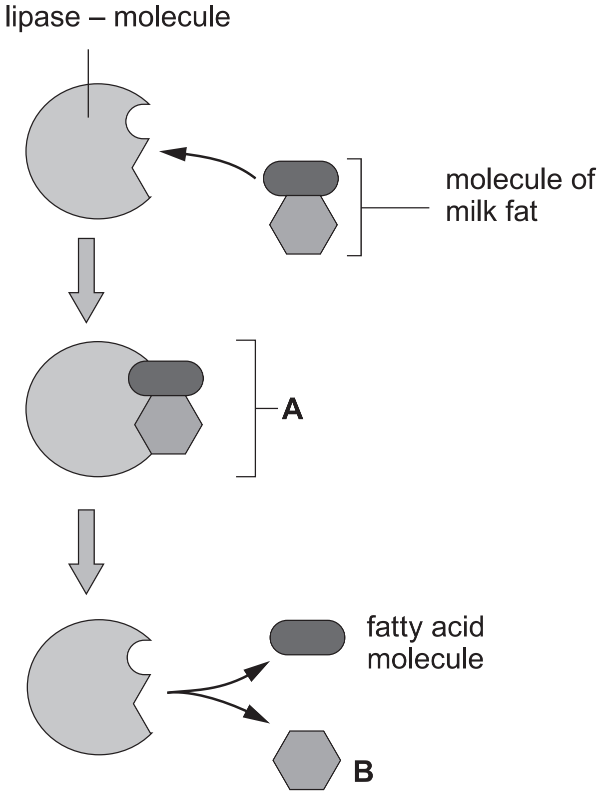
5. Describe the pathway taken by the blood in a double circulatory system. Start your account with the return of blood along the pulmonary vein from the lungs to the left atrium of the heart. Your account must include details of how backflow of blood is prevented. [6 QER]

6

Examiner only

6. **Image 6.1** represents the lock and key model of enzyme action involving lipase and milk fat.

Image 6.1



(a) State the name of:

(i) structure **A**; [1]

....

(ii) molecule **B**. [1]

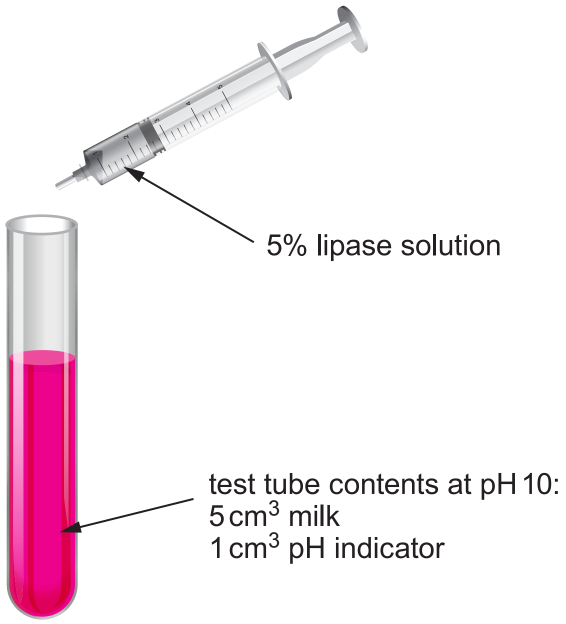
....

Examiner only

(b) A class of students investigated how lipase activity changes with temperature.
They used a pH indicator that is pink in alkaline solutions of about pH 10. When the pH drops below pH 8 it goes colourless.
A solution of full-fat milk, lipase and indicator at pH 10 will change from pink to colourless as the **fat** in milk is broken down producing **fatty acids**. This reduces the pH to below 8. The time taken for this reaction to occur is affected by temperature.

The students worked in pairs and set up the apparatus as shown in **Image 6.2**.

Image 6.2



Each pair of students investigated a different temperature.

- The test tube and syringe were placed in a water bath.
- At 5 minutes, 1 cm^3 of lipase solution from the syringe was added to the test tube.
- The time taken for the solution in the test tube to change from pink to colourless was recorded.
- The experiment was carried out three times for each temperature.

The class results are shown in **Table 6.3**.

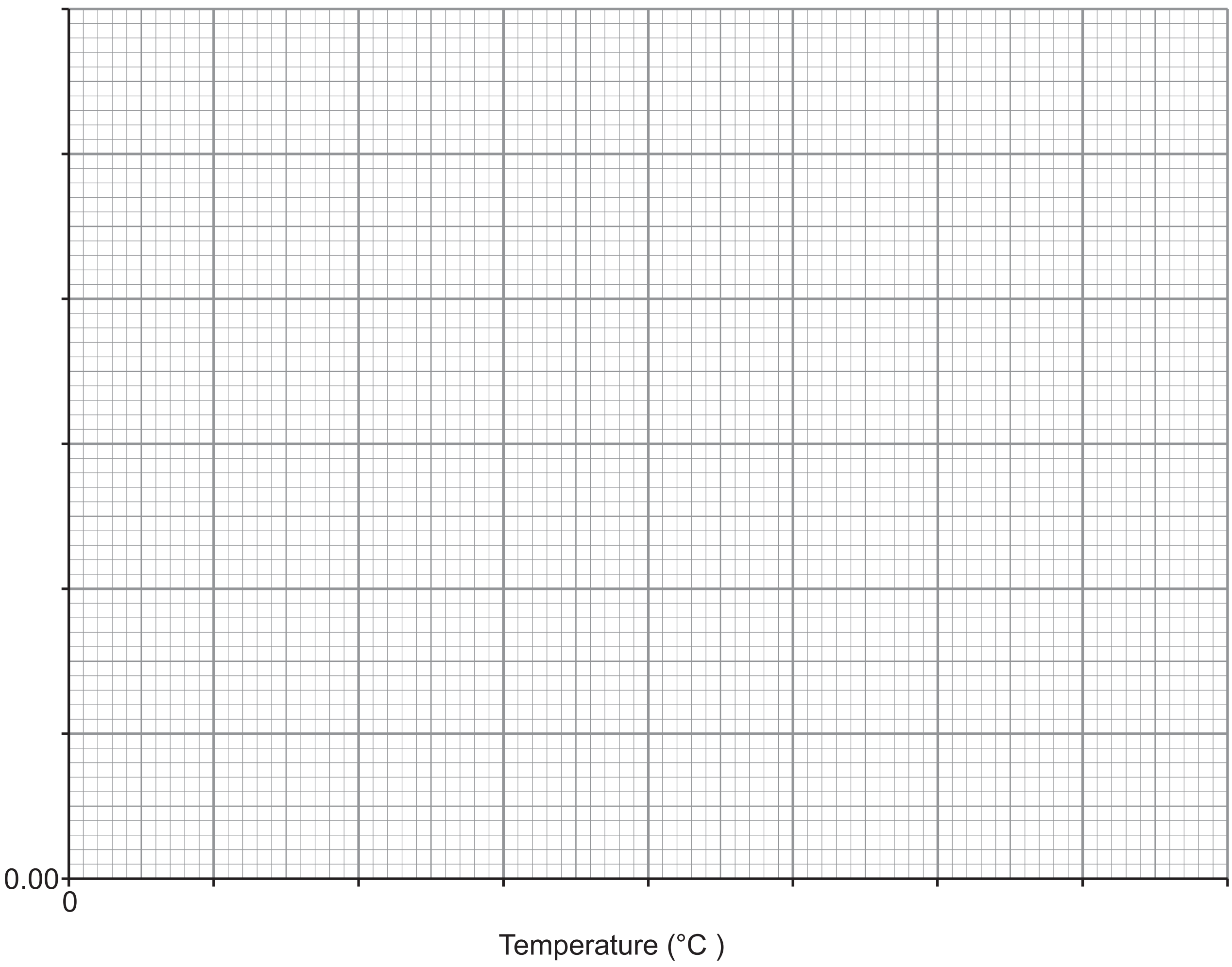
Table 6.3.

	Time for indicator to become colourless (s)				Rate of reaction 1 ÷ mean time (per second)
Temperature (°C)	Trial 1	Trial 2	Trial 3	Mean	
0 (ice bath)	no change	no change	no change	no change	0.00
20	8.0	6.0	7.0	7.00	0.14
40	5.0	4.0	5.0	4.67	0.21
60	10.0	9.0	9.0	9.33	....
80	no change	no change	no change	no change	0.00

Examiner only

(i) The mean time for each temperature can be converted to rate of reaction by calculating 1 ÷ mean time.
Complete Table 6.3 by calculating the rate of reaction at 60 °C. **Give your answer to 2 decimal places**. [1]
Space for working.

(ii) On the grid below, draw a **line graph** of **rate of reaction** against temperature. You should join the plots with a ruler. [4]



(iii) Describe the effect of temperature on the rate of reaction of lipase. [1]

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Examiner only

(iv) Explain the effect of temperature on the rate of reaction between 0°C and 40°C. [2]

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(c) State why the test tube and syringe were left in the water bath for 5 minutes before the lipase solution was added to the test tube. [1]

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(d) Suggest why skimmed milk could not be used in this experiment. [1]

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12

Examiner only

7. (a) **Complete the following table** about the processes by which substances move through cell membranes. [3]

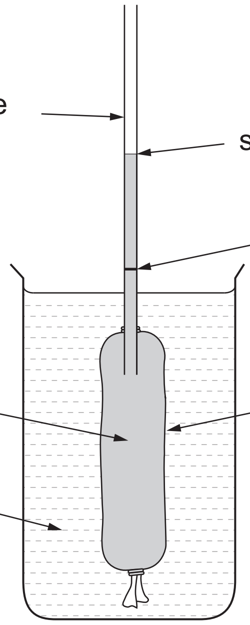
Place a tick (✓) or a cross (×) in each box to indicate if the statement applies to each process or not.

	Active transport	Osmosis	Diffusion
Energy (ATP) needed	....	....	....
Against a concentration gradient	....	....	....
Down a concentration gradient	....	....	....

Examiner only

(b) The apparatus shown in **Image 7.1** was set up using a piece of Visking tubing filled with sucrose solution. The Visking tubing was knotted at its bottom end and tied at its top end to a length of glass tube. The Visking tubing was then placed in a beaker of water and left for 1 hour. At the end of this time the sucrose solution had risen up the tube.

Image 7.1



Explain why the sucrose solution moved up the glass tube. [4]

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7

END OF PAPER

 (3430UA0-1) **Turn over.**

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

S23-3430UA0-1

3430UA0-1

TUESDAY, 13 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 1: BIOLOGY 1
HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**10**	
3.	**7**	
4.	**10**	
5.	**10**	
6.	**10**	
7.	**8**	
Total	**60**	

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ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.

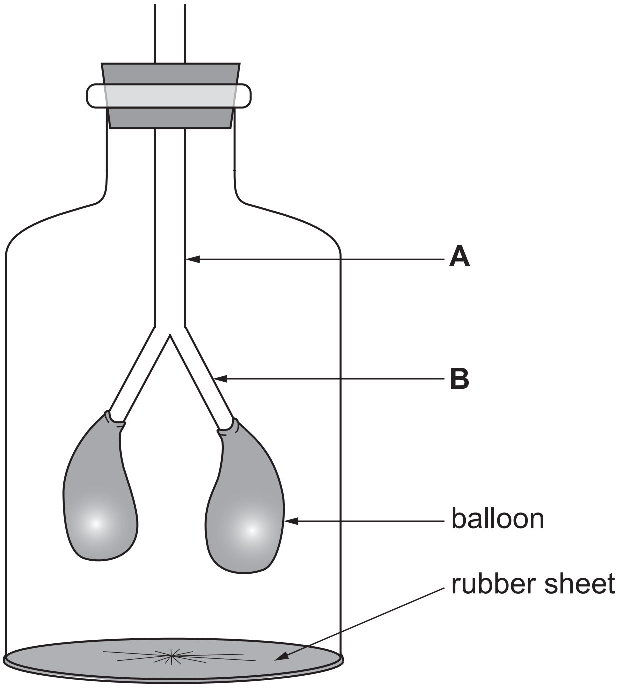
 VP/GR*(S20-3430UA0-1)

Examiner only

Answer **all** questions.

1. **Image 1** shows the bell jar model which is being used to demonstrate inspiration.

Image 1



(a) (i) Parts **A** and **B** in **Image 1** represent two parts of the human respiratory system.

Complete the following sentence: [2]

In the bell jar model, part **A** represents the .... and part **B** represents the ....

(ii) As the rubber sheet is pulled down, the air pressure inside the bell jar changes. Explain how the air pressure change in the bell jar causes the balloons to inflate. [2]

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Examiner only

(b) The balloons are in an air-tight space in the bell jar. Lungs are sealed in an air-tight space in the chest.

A wound to the chest means the chest is no longer air-tight. As a result, the wounded person cannot fully inflate their lungs.

Explain why a person with a hole in the chest cannot fully inflate their lungs. [1]

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3430UA01 03

Examiner only

2. **Image 2.1** and **Image 2.2** show examples of two methods of farming hens for egg production.

Image 2.1 – Intensively farmed chickens

Image 2.2 – Free-range chickens

- Intensive farming methods maximise production by controlling the conditions in which farm animals are kept. Inside animal sheds, temperatures may be kept high and each animal is given limited space.
- In 2015, a survey of 2000 people (*OnePoll*) found that 80% of those questioned always or often bought free-range eggs, even though they were more expensive.

Table 2.3 shows egg production in the UK by intensive and free-range methods between 2006 and 2016.

Table 2.3

Year	Egg production (billion)	
	intensive	free-range
2006	4.1	1.9
2008	4.0	2.1
2010	3.8	2.8
2012	3.5	3.4
2014	3.9	3.3
2016	3.8	3.8

(a) Use **Table 2.3** and the information above to answer the following questions.

(i) Calculate the percentage increase in the production of **free-range eggs** between 2006 and 2016. [2]

Increase = ... %

Examiner only

(ii) Give **one** way that intensive farming methods minimise energy loss from farm animals. Explain your answer. [2]

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(b) Suggest **two** reasons why many people buy free-range eggs, even though they are more expensive than eggs produced by intensive farming methods. [2]

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(c) (i) State **one** possible **pollutant** from intensive farming methods and explain how it could damage the environment. [2]

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....

(ii) Farmers in Wales who plan to develop intensive methods of food production on rural land must first submit their plans to biologists at Natural Resources Wales. Describe the role of the biologists at both the planning stage and when the intensive farm is operating fully. [2]

Planning stage

....

....

When operating fully

....

....

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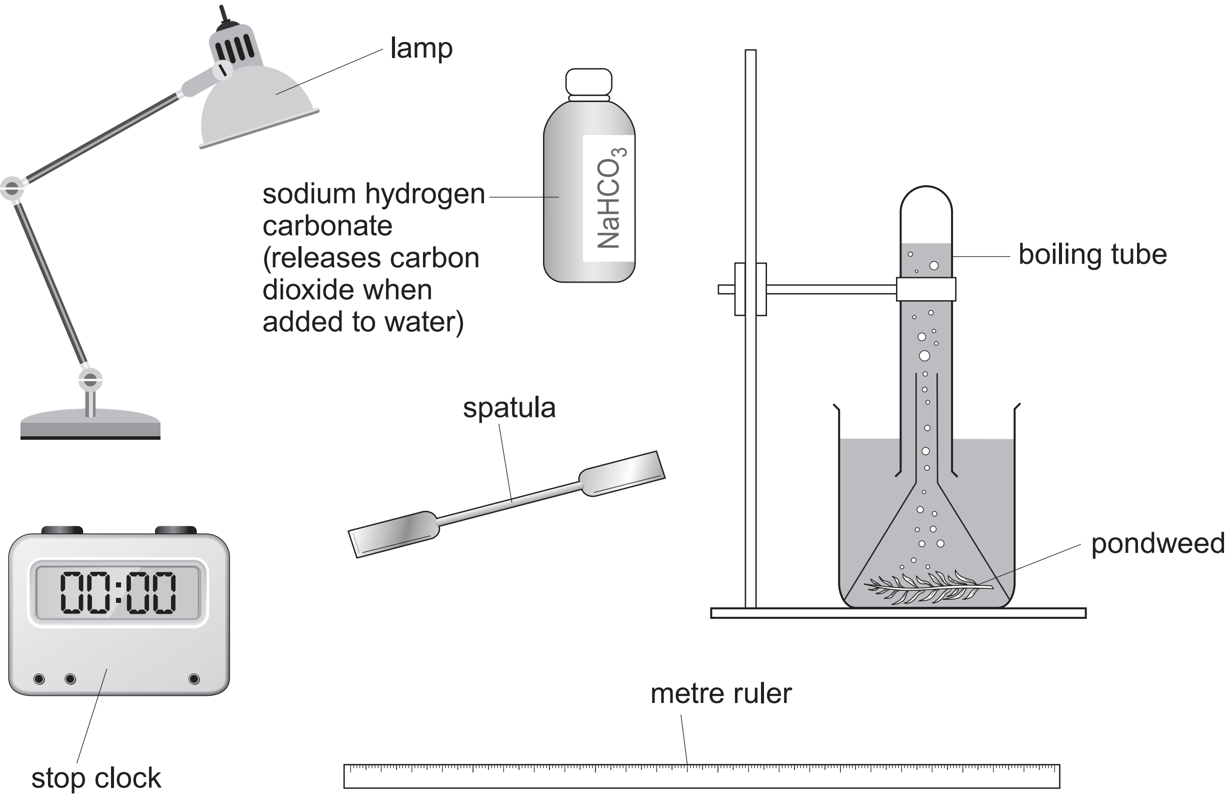
Examiner only

3. (a) Describe how you would investigate the effect of light intensity on the rate of photosynthesis using **only** the apparatus shown in **Image 3**. [6 QER]

Image 3



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Examiner only

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(b) It is possible to determine if photosynthesis is taking place by testing a leaf for the presence of starch using iodine solution. Explain why this test is not suitable for determining the **rate** of photosynthesis. [1]

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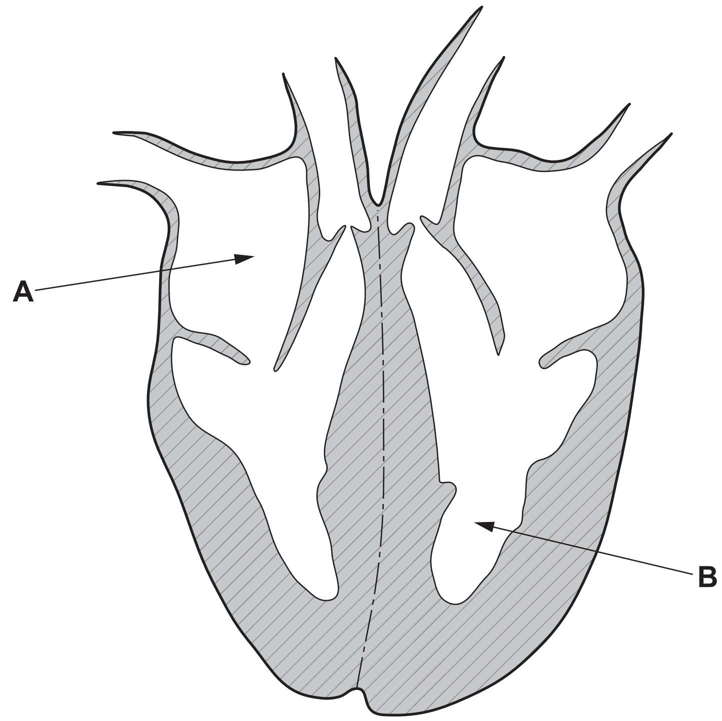
Examiner only

4. **Image 4.1** shows a section through a human heart.

Image 4.1



(a) (i) State the name of chambers **A** and **B** shown in **Image 4.1**. [1]

A ....

B ....

(ii) Humans rely on a double circulatory system. State what is meant by a double circulatory system. [2]

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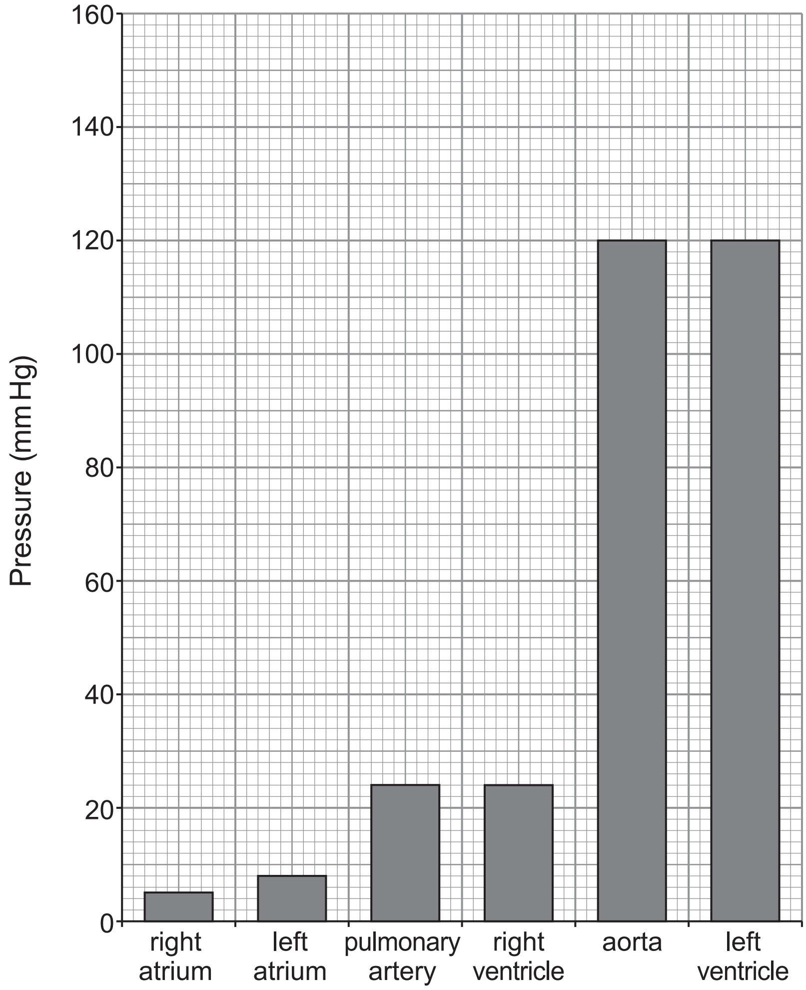
Examiner only

(b) **Graph 4.2** shows the maximum pressure that occurs in the heart and associated blood vessels during the cardiac cycle.

Graph 4.2



(i) I. Calculate the difference in the maximum pressure between the right ventricle and the left ventricle. [1]

Difference in pressure = ... mm Hg

II. State how the structure of the left ventricle causes this difference in pressure. [1]

....

....

(ii) Suggest why the maximum pressure in the aorta is the same as that in the left ventricle. [1]

....

....

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(c) Cardiovascular disease (CVD) is a general term for conditions affecting the heart or blood vessels. High blood pressure is one of the most important risk factors for CVD. Scientists carried out an investigation to test the following hypothesis:

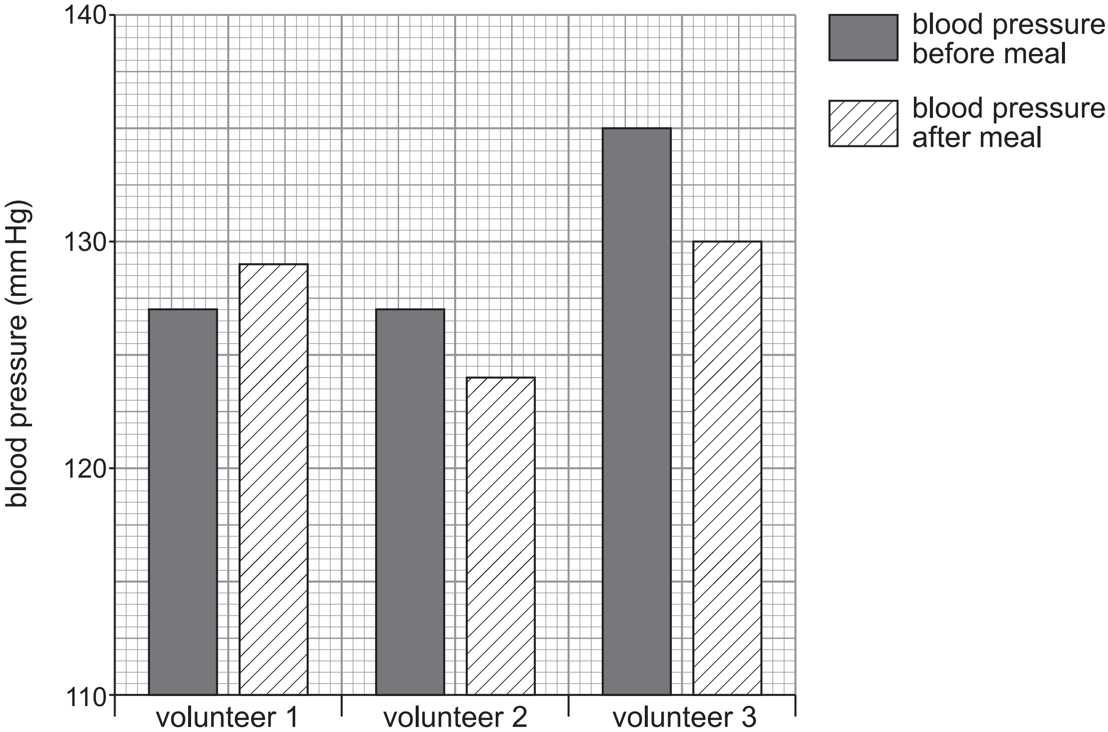
'Eating food or drink that is high in nitrate causes blood vessels to widen, leading to a decrease in blood pressure.'

The investigation was carried out as follows:

- Three volunteers were each given a different meal.
- Volunteer **1**: A salad made from vegetables that naturally contain almost no nitrate.
- Volunteer **2**: A salad containing spinach and rocket, which have naturally high levels of nitrate.
- Volunteer **3**: A portion of beetroot juice, calculated to contain exactly the same quantity of nitrate as the meal eaten by volunteer **2**.
- The blood pressure of each volunteer was tested before and after their meal.

The results are shown in **Graph 4.3**.

Graph 4.3



Examiner only

State **two** conclusions which can be made from these results. [2]

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(d) Suggest why the widening of coronary arteries can lower the risk of a heart attack. [2]

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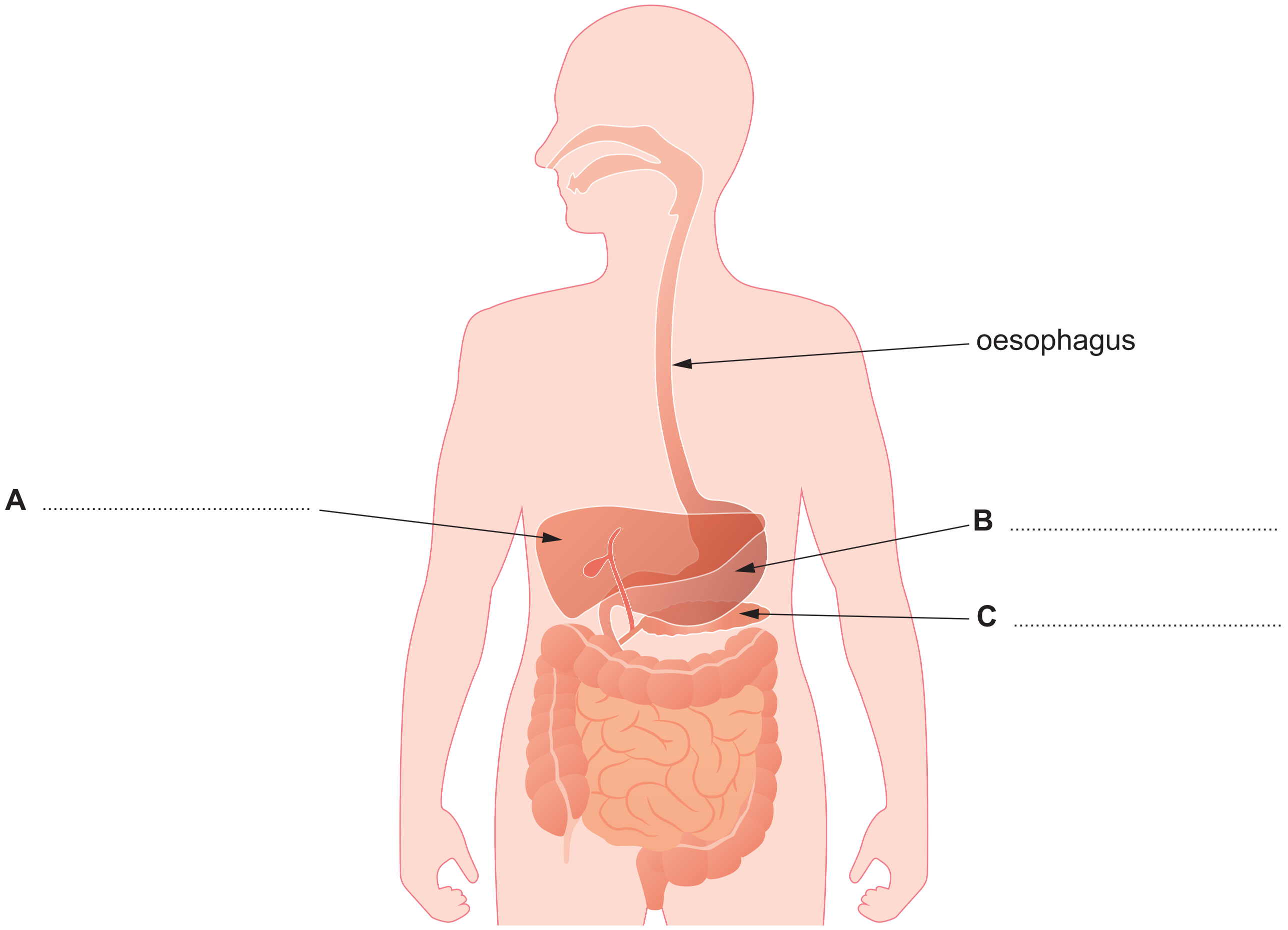
Examiner only

5. **Image 5.1** shows part of the digestive system.

Image 5.1



(a) **Label parts A** to **C** on **Image 5.1**. [2]

(b) Explain why carbohydrase cannot digest fat. [1]

....

....

Examiner only

(c) Scientists investigated absorption of three different sugars in a sample of healthy small intestine in the laboratory. They analysed the absorption of the sugars in the presence or absence of cyanide. Cyanide is a chemical that prevents respiration taking place in cells.

The results are shown in **Table 5.2**.

Table 5.2

Sugar	Absorption without cyanide (a.u.)	Absorption with cyanide (a.u.)
glucose	100	33
xylose	30	30
arabinose	29	29

Use the information in **Table 5.2** to answer the following questions.

(i) State which sugar is absorbed by active transport. Explain your answer. [4]

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(ii) One of the scientists stated that all three of the sugars in the investigation could be absorbed by diffusion. Explain how the evidence in **Table 5.2** supports this conclusion. [2]

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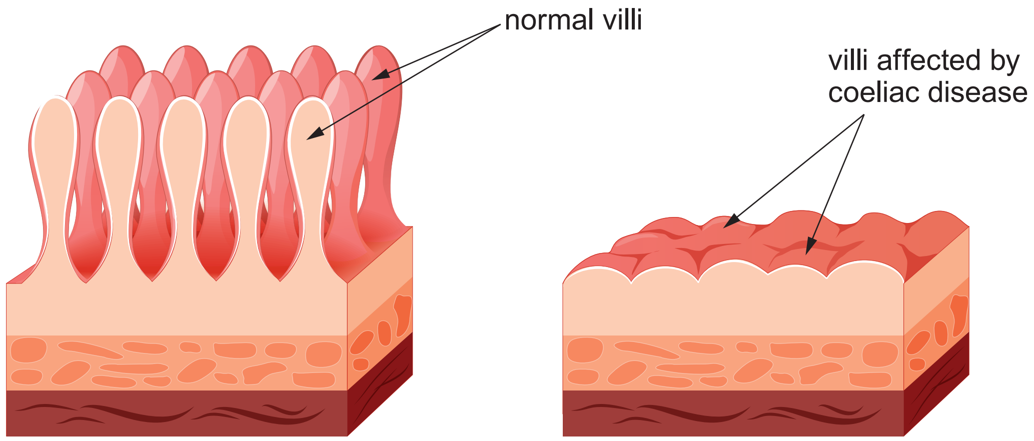
Examiner only

(d) **Image 5.3** shows the structure of the lining of the small intestine of a healthy individual and an individual with coeliac disease. Coeliac disease causes a flattening of the villi lining the small intestine.

Image 5.3



Use **Image 5.3** to explain why individuals who have coeliac disease are unable to fully absorb their digested food. [1]

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Examiner only

6. **Image 6** appeared on the BBC news website in 2018.

Image 6

Florida powerless to stop killer algal bloom.

In Florida, the uncontrolled growth of toxic algae known as "red tide" has caused an emergency. The sea waters are now a dark red colour. The red tide has left tonnes of dead animals, affected local businesses and worried the inhabitants. The abundance of the toxic algae has devastating effects all the way along food chains. More than 5 tonnes of dead fish have been reported, and the death of many dolphins. Shellfish such as clams, oysters and mussels are not affected but have been found to contain toxins produced by the algae.

Red tide

(a) Use the information in **Image 6** to answer the following questions.

(i) Red tide has been suspected of causing poisoning in humans. Whilst the poisonings have not been fatal many individuals have been admitted to hospital.

Suggest how humans were poisoned. [2]

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Examiner only

(ii) The recent red tide has killed approximately 500 turtles, as they ingest the algae when they feed on sea grasses. Prior to this, the population of turtles in Florida waters was estimated at 51 000. Calculate the percentage of turtles that have died as a result of red tide. **Give your answer to two significant figures.** [3]

Percentage of turtles killed = ... %

(b) Dead zones are areas of water with reduced oxygen levels. Some scientists have argued that the red tide can lead to dead zones forming in the sea. Explain how the increase in algal growth described in the article could lead to dead zones. [5]

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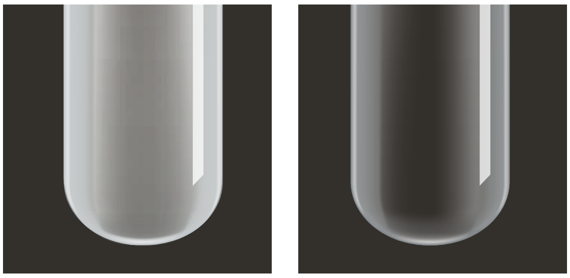
TURN OVER FOR QUESTION 7

Examiner only

7. Dried milk powder and whey powder both contain protein. A weight lifter wanted to determine which would be more beneficial in his diet for his performance.

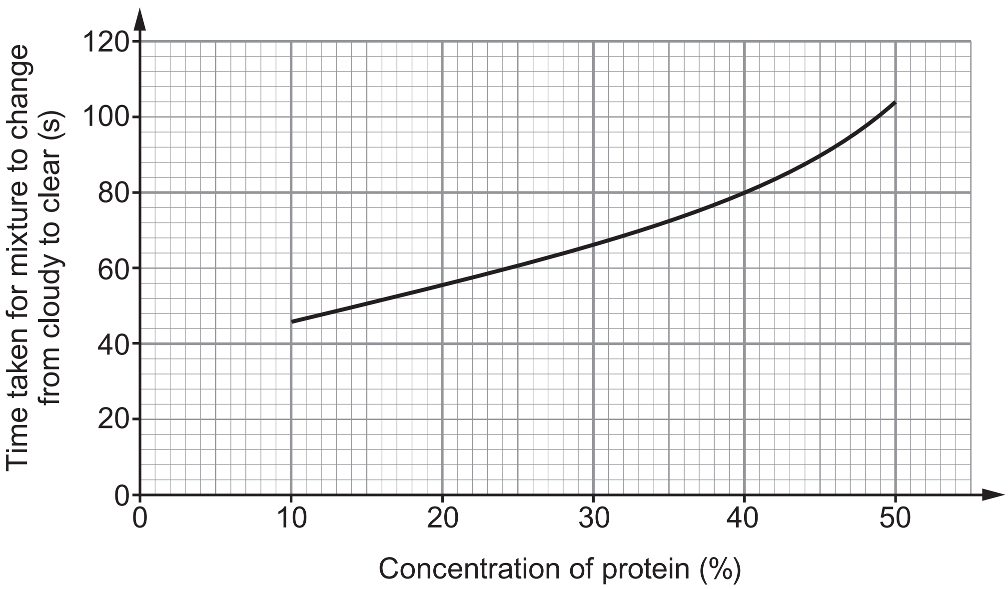
In order to do this, he first prepared equal volumes of known concentrations of protein for comparison. He then added $5\,cm^3$ of 5% enzyme solution to each and recorded the time taken for the protein-enzyme mixture to go from cloudy to clear. **This colour change is shown in Image 7.1.**

Image 7.1



His results are shown in the **Graph 7.2**.

Graph 7.2



(a) (i) State the independent and dependent variables. [2]

Independent variable ....

Dependent variable ....

Examiner only

(ii) Name the type of enzyme which was added to the protein **and** explain the change in appearance from cloudy to clear. [2]

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(iii) State **one** source of uncertainty in the experimental method **and** suggest how it could be overcome. [2]

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(b) The weight lifter then carried out the experiment using a sample of dried milk powder followed by a sample of whey powder. The milk powder took 72 seconds to clear and the whey powder took 100 seconds to clear.

Use data from **Graph 7.2** to explain why the weight lifter concluded he should use whey powder rather than dried milk powder in his diet to improve his performance. [2]

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END OF PAPER